Results of Experiments

Note: This file contains training results on training data and prediction results on test data concerning the experiments as described in chapter 7 of the master thesis. **In this file the results of the CNN-model are documented only.** Each experiment is conducted three times with different sampled training and validation data as well as with different initial model weights[1]. The sub experiment with different training/validation/initial model weight sets are marked with “random_state” labels 42, 31, 7.

The listed confusion matrices show the ground truth label on the left side, the predicted label on the top.

Experiment	Description
1a - f	Class 0 = all other, class 1 = SR, different sequence length of 100, 800, 600, 400, 200 data points and randomly mixed sequence length (1f)
2a - f	Class 0 = all other, class 1 = SR, different sequence length of 100, 800, 600, 400, 200 data points and randomly mixed sequence length (1f) With augmented data according to Chapter "Data augmentation" of master thesis
3a, 3 b	Class 0 = all other, class 1 = SR, pretraining of model weights by inclusion of handwritten text snipes in the training (3a) and fine training without handwritten text snippets (3b).
4a-c	Classification results of different label scenarios 4a: class 0 = SR, class 1 = all other, class 2 = binary label, class 3 = single label 4b: class 0 = SR, class 1 = all other, class 2 = form, class 3 = rhythm 4c: class 0 = 'ABQRS', class 1= 'AFIB', class 2 = 'INVT', class 3 = 'LNGQT', class 4 = 'LPR', class 5 = 'LVOLT', class 6 = 'NDT', class 7 = 'NST_' class 8 = 'NT_', class 9 = 'PAC', class 10 = 'PACE', class 11 = 'PVC', class 12 ='QWAVE', class 13 = 'SARRH', class 14 = 'SBRAD', class 15 = 'SR', class 16 = 'STACH', class 17 = 'STD_', class 18 = 'SVARR', class 19 = 'VCLVH', class 20 = 'all_other'
5	Multilabel classification Label: SR, ABQRS, AFIB, all other Resulting eight label combinations: SR, ABQRS, AFIB, all other, ABQRS + all_other, AFIB + all_other, ABQRS + AFIB, ABQRS + AFIB + all_other

[1] According to different starting conditions by different sampled training data and initial model weights of the training the results differ slightly between "random_state" 42, 31 and 7.

Experiment 1a / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5300
Accuracy: 0.7385
F1-score: 0.7550
Confusion matrix:
   0    1
0[4823 2721]
1[1039 5795]

Training classification report:
              precision    recall  f1-score       support
class_0        0.822757  0.639316  0.719529   7544.000000
class_1        0.680484  0.847966  0.755049   6834.000000
accuracy       0.738489  0.738489  0.738489      0.738489
macro avg      0.751620  0.743641  0.737289  14378.000000
weighted avg   0.755133  0.738489  0.736412  14378.000000

Test metrics:
Loss:     0.5811
Accuracy: 0.7004
F1-score: 0.7260
Confusion matrix:
   0   1
0[645 468]
1[169 844]

Test classification report:
              precision    recall  f1-score      support
class_0        0.792383  0.579515  0.669434  1113.000000
class_1        0.643293  0.833169  0.726022  1013.000000
accuracy       0.700376  0.700376  0.700376     0.700376
macro avg      0.717838  0.706342  0.697728  2126.000000
weighted avg   0.721344  0.700376  0.696397  2126.000000
```

Experiment 1a / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4277
Accuracy: 0.8025
F1-score: 0.8019
Confusion matrix:
   0    1
0[5789 1755]
1[1085 5749]

Training classification report:
              precision    recall  f1-score       support
class_0        0.842159  0.767365  0.803024   7544.000000
class_1        0.766125  0.841235  0.801925   6834.000000
accuracy       0.802476  0.802476  0.802476      0.802476
macro avg      0.804142  0.804300  0.802474  14378.000000
weighted avg   0.806019  0.802476  0.802502  14378.000000


Test metrics:
Loss:     0.5500
Accuracy: 0.7244
F1-score: 0.7272
Confusion matrix:
  0    1
0[759 354]
1[232 781]

Test classification report:
              precision    recall  f1-score      support
class_0        0.765893  0.681941  0.721483  1113.000000
class_1        0.688106  0.770977  0.727188  1013.000000
accuracy       0.724365  0.724365  0.724365     0.724365
macro avg      0.726999  0.726459  0.724335  2126.000000
weighted avg   0.728829  0.724365  0.724201  2126.000000
```

Experiment 1a / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4569
Accuracy: 0.7891
F1-score: 0.7848
Confusion matrix:
   0    1
0[5814 1730]
1[1303 5531]

Training classification report:
              precision    recall  f1-score       support
class_0        0.816917  0.770679  0.793125   7544.000000
class_1        0.761741  0.809336  0.784817   6834.000000
accuracy       0.789053  0.789053  0.789053      0.789053
macro avg      0.789329  0.790007  0.788971  14378.000000
weighted avg   0.790691  0.789053  0.789176  14378.000000


Test metrics:
Loss:     0.5384
Accuracy: 0.7291
F1-score: 0.7280
Confusion matrix:
  0    1
0[779 334]
1[242 771]
```

Experiment 1b / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5366
Accuracy: 0.7348
F1-score: 0.7530
Confusion matrix:
  0      1
0[4754 2790]
1[1023 5811]

Training classification report:
              precision    recall  f1-score       support
class_0        0.822918  0.630170  0.713760   7544.000000
class_1        0.675619  0.850307  0.752964   6834.000000
accuracy       0.734803  0.734803  0.734803      0.734803
macro avg      0.749269  0.740238  0.733362  14378.000000
weighted avg   0.752906  0.734803  0.732394  14378.000000


Test metrics:
Loss:     0.5884
Accuracy: 0.6976
F1-score: 0.7249
Confusion matrix:
  0    1
0[636 477]
1[166 847]

Test classification report:
              precision    recall  f1-score      support
class_0        0.793017  0.571429  0.664230  1113.000000
class_1        0.639728  0.836130  0.724861  1013.000000
accuracy       0.697554  0.697554  0.697554     0.697554
macro avg      0.716373  0.703779  0.694545  2126.000000
weighted avg   0.719978  0.697554  0.693119  2126.000000
```

Experiment 1b / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4243
Accuracy: 0.8055
F1-score: 0.8063
Confusion matrix:
  0     1
0[5760 1784]
1[1013 5821]

Training classification report:
              precision    recall  f1-score       support
class_0        0.850436  0.763521  0.804638   7544.000000
class_1        0.765417  0.851771  0.806289   6834.000000
accuracy       0.805467  0.805467  0.805467      0.805467
macro avg      0.807927  0.807646  0.805463  14378.000000
weighted avg   0.810026  0.805467  0.805422  14378.000000


Test metrics:
Loss:     0.5608
Accuracy: 0.7168
F1-score: 0.7236
Confusion matrix:
  0    1
0[736 377]
1[225 788]

Test classification report:
              precision    recall  f1-score      support
class_0        0.765869  0.661276  0.709740  1113.000000
class_1        0.676395  0.777887  0.723600  1013.000000
accuracy       0.716839  0.716839  0.716839     0.716839
macro avg      0.721132  0.719582  0.716670  2126.000000
weighted avg   0.723236  0.716839  0.716344  2126.000000
```

Experiment 1c / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4628
Accuracy: 0.7892
F1-score: 0.7897
Confusion matrix:
  0     1
0[5656 1888]
1[1143 5691]

Training classification report:
              precision    recall  f1-score       support
class_0        0.831887  0.749735  0.788677   7544.000000
class_1        0.750891  0.832748  0.789704   6834.000000
accuracy       0.789192  0.789192  0.789192      0.789192
macro avg      0.791389  0.791241  0.789191  14378.000000
weighted avg   0.793389  0.789192  0.789165  14378.000000


Test metrics:
Loss:     0.5457
Accuracy: 0.7225
F1-score: 0.7299
Confusion matrix:
  0    1
0[739 374]
1[216 797]

Test classification report:
              precision    recall  f1-score      support
class_0        0.773822  0.663971  0.714700  1113.000000
class_1        0.680615  0.786772  0.729853  1013.000000
accuracy       0.722484  0.722484  0.722484     0.722484
macro avg      0.727218  0.725372  0.722277  2126.000000
weighted avg   0.729411  0.722484  0.721920  2126.000000
```

Experiment 1c / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5516
Accuracy: 0.7255
F1-score: 0.7517
Confusion matrix:
   0    1
0[4455 3089]
1[ 858 5976]

Training classification report:
              precision    recall  f1-score       support
class_0        0.838509  0.590536  0.693008   7544.000000
class_1        0.659239  0.874451  0.751745   6834.000000
accuracy       0.725483  0.725483  0.725483      0.725483
macro avg      0.748874  0.732493  0.722377  14378.000000
weighted avg   0.753300  0.725483  0.720926  14378.000000


Test metrics:
Loss:     0.6020
Accuracy: 0.6947
F1-score: 0.7281
Confusion matrix:
  0    1
0[608 505]
1[144 869]

Test classification report:
              precision    recall  f1-score      support
class_0        0.808511  0.546271  0.652011  1113.000000
class_1        0.632460  0.857848  0.728111  1013.000000
accuracy       0.694732  0.694732  0.694732     0.694732
macro avg      0.720485  0.702060  0.690061  2126.000000
weighted avg   0.724626  0.694732  0.688271  2126.000000
```

Experiment 1c / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4307
Accuracy: 0.8017
F1-score: 0.8018
Confusion matrix:
   0     1
0[5760 1784]
1[1067 5767]

Training classification report:
              precision    recall  f1-score       support
class_0        0.843709  0.763521  0.801614   7544.000000
class_1        0.763740  0.843869  0.801807   6834.000000
accuracy       0.801711  0.801711  0.801711      0.801711
macro avg      0.803724  0.803695  0.801711  14378.000000
weighted avg   0.805699  0.801711  0.801706  14378.000000


Test metrics:
Loss:     0.5601
Accuracy: 0.7150
F1-score: 0.7197
Confusion matrix:
  0    1
0[742 371]
1[235 778]

Test classification report:
              precision    recall  f1-score      support
class_0        0.759468  0.666667  0.710048  1113.000000
class_1        0.677111  0.768016  0.719704  1013.000000
accuracy       0.714958  0.714958  0.714958     0.714958
macro avg      0.718289  0.717341  0.714876  2126.000000
weighted avg   0.720226  0.714958  0.714649  2126.000000
```

Experiment 1c / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4717
Accuracy: 0.7806
F1-score: 0.7845
Confusion matrix:
   0    1
0[5483 2061]
1[1093 5741]

Training classification report:
              precision    recall  f1-score       support
class_0        0.833790  0.726803  0.776629   7544.000000
class_1        0.735837  0.840064  0.784504   6834.000000
accuracy       0.780637  0.780637  0.780637      0.780637
macro avg      0.784813  0.783434  0.780566  14378.000000
weighted avg   0.787232  0.780637  0.780372  14378.000000


Test metrics:
Loss:     0.5537
Accuracy: 0.7173
F1-score: 0.7269
Confusion matrix:
   0   1
0[725 388]
1[213 800]

Test classification report:
              precision    recall  f1-score     support
class_0        0.772921  0.651393  0.706972  1113.00000
class_1        0.673401  0.789733  0.726942  1013.00000
accuracy       0.717310  0.717310  0.717310     0.71731
macro avg      0.723161  0.720563  0.716957  2126.00000
weighted avg   0.725501  0.717310  0.716488  2126.00000
```

Experiment 1d / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5295
Accuracy: 0.7418
F1-score: 0.7580
Confusion matrix:
    0    1
0[4854 2690]
1[1022 5812]

Training classification report:
              precision    recall  f1-score       support
class_0        0.826072  0.643425  0.723398   7544.000000
class_1        0.683604  0.850454  0.757955   6834.000000
accuracy       0.741828  0.741828  0.741828      0.741828
macro avg      0.754838  0.746939  0.740677  14378.000000
weighted avg   0.758356  0.741828  0.739823  14378.000000


Test metrics:
Loss:     0.5938
Accuracy: 0.6999
F1-score: 0.7240
Confusion matrix:
   0   1
0[651 462]
1[176 837]

Test classification report:
              precision    recall  f1-score      support
class_0        0.787183  0.584906  0.671134  1113.000000
class_1        0.644342  0.826259  0.724048  1013.000000
accuracy       0.699906  0.699906  0.699906     0.699906
macro avg      0.715762  0.705582  0.697591  2126.000000
weighted avg   0.719122  0.699906  0.696347  2126.000000
```

Experiment 1d / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.3902
Accuracy: 0.8297
F1-score: 0.8325
Confusion matrix:
   0    1
0[5842 1702]
1[ 747 6087]

Training classification report:
              precision    recall  f1-score      support
class_0        0.886629  0.774390  0.826718   7544.00000
class_1        0.781487  0.890694  0.832524   6834.00000
accuracy       0.829670  0.829670  0.829670      0.82967
macro avg      0.834058  0.832542  0.829621  14378.00000
weighted avg   0.836654  0.829670  0.829477  14378.00000


Test metrics:
Loss:     0.5808
Accuracy: 0.7117
F1-score: 0.7217
Confusion matrix:
  0    1
0[718 395]
1[218 795]

Test classification report:
              precision    recall  f1-score      support
class_0        0.767094  0.645103  0.700830  1113.000000
class_1        0.668067  0.784798  0.721743  1013.000000
accuracy       0.711665  0.711665  0.711665     0.711665
macro avg      0.717581  0.714950  0.711286  2126.000000
weighted avg   0.719910  0.711665  0.710795  2126.000000
```

Experiment 1d / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4680
Accuracy: 0.7848
F1-score: 0.7783
Confusion matrix:
   0     1
0[5853 1691]
1[1403 5431]

Training classification report:
              precision    recall  f1-score      support
class_0        0.806643  0.775848  0.790946   7544.00000
class_1        0.762567  0.794703  0.778303   6834.00000
accuracy       0.784810  0.784810  0.784810      0.78481
macro avg      0.784605  0.785276  0.784625  14378.00000
weighted avg   0.785693  0.784810  0.784937  14378.00000


Test metrics:
Loss:     0.5559
Accuracy: 0.7234
F1-score: 0.7234
Confusion matrix:
   0   1
0[769 344]
1[244 769]

Test classification report:
              precision    recall  f1-score      support
class_0        0.759131  0.690925  0.723424  1113.000000
class_1        0.690925  0.759131  0.723424  1013.000000
accuracy       0.723424  0.723424  0.723424     0.723424
macro avg      0.725028  0.725028  0.723424  2126.000000
weighted avg   0.726632  0.723424  0.723424  2126.000000
```

Experiment 1e / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5558
Accuracy: 0.7249
F1-score: 0.7352
Confusion matrix:
    0    1
0[4929 2615]
1[1341 5493]

Training classification report:
              precision    recall  f1-score       support
class_0        0.786124  0.653367  0.713624   7544.000000
class_1        0.677479  0.803775  0.735243   6834.000000
accuracy       0.724857  0.724857  0.724857      0.724857
macro avg      0.731802  0.728571  0.724433  14378.000000
weighted avg   0.734484  0.724857  0.723900  14378.000000


Test metrics:
Loss:     0.6103
Accuracy: 0.6825
F1-score: 0.7017
Confusion matrix:
   0   1
0[657 456]
1[219 794]

Test classification report:
              precision    recall  f1-score      support
class_0        0.750000  0.590296  0.660633  1113.000000
class_1        0.635200  0.783810  0.701723  1013.000000
accuracy       0.682502  0.682502  0.682502     0.682502
macro avg      0.692600  0.687053  0.681178  2126.000000
weighted avg   0.695300  0.682502  0.680212  2126.000000
```

Experiment 1e / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5101
Accuracy: 0.7532
F1-score: 0.7534
Confusion matrix:
   0    1
0[5411 2133]
1[1415 5419]

Training classification report:
              precision    recall  f1-score       support
class_0        0.792704  0.717259  0.753097   7544.000000
class_1        0.717558  0.792947  0.753371   6834.000000
accuracy       0.753234  0.753234  0.753234      0.753234
macro avg      0.755131  0.755103  0.753234  14378.000000
weighted avg   0.756987  0.753234  0.753227  14378.000000



Test metrics:
Loss:     0.5908
Accuracy: 0.6914
F1-score: 0.6994
Confusion matrix:
   0   1
0[707 406]
1[250 763]

Test classification report:
              precision    recall  f1-score      support
class_0        0.738767  0.635220  0.683092  1113.000000
class_1        0.652695  0.753208  0.699358  1013.000000
accuracy       0.691439  0.691439  0.691439     0.691439
macro avg      0.695731  0.694214  0.691225  2126.000000
weighted avg   0.697755  0.691439  0.690843  2126.000000
```

Experiment 1e / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4687
Accuracy: 0.7764
F1-score: 0.7548
Confusion matrix:
  0     1
0[6214 1330]
1[1885 4949]

Training classification report:
              precision    recall  f1-score       support
class_0        0.767255  0.823701  0.794477   7544.000000
class_1        0.788183  0.724173  0.754823   6834.000000
accuracy       0.776394  0.776394  0.776394      0.776394
macro avg      0.777719  0.773937  0.774650  14378.000000
weighted avg   0.777202  0.776394  0.775629  14378.000000


Test metrics:
Loss:     0.5962
Accuracy: 0.6867
F1-score: 0.6616
Confusion matrix:
   0   1
0[809 304]
1[362 651]

Test classification report:
              precision    recall  f1-score      support
class_0        0.690863  0.726864  0.708406  1113.000000
class_1        0.681675  0.642646  0.661585  1013.000000
accuracy       0.686736  0.686736  0.686736     0.686736
macro avg      0.686269  0.684755  0.684996  2126.000000
weighted avg   0.686485  0.686736  0.686097  2126.000000
```

Experiment 1f / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5498
Accuracy: 0.7246
F1-score: 0.7360
Confusion matrix:
   0     1
0[4897 2647]
1[1313 5521]

Training classification report:
              precision    recall  f1-score       support
class_0        0.788567  0.649125  0.712084   7544.000000
class_1        0.675930  0.807872  0.736035   6834.000000
accuracy       0.724579  0.724579  0.724579      0.724579
macro avg      0.732249  0.728499  0.724059  14378.000000
weighted avg   0.735030  0.724579  0.723468  14378.000000


Test metrics:
Loss:     0.5925
Accuracy: 0.6943
F1-score: 0.7149
Confusion matrix:
   0   1
0[661 452]
1[198 815]

Test classification report:
              precision    recall  f1-score      support
class_0        0.769499  0.593890  0.670385  1113.000000
class_1        0.643252  0.804541  0.714912  1013.000000
accuracy       0.694262  0.694262  0.694262     0.694262
macro avg      0.706376  0.699216  0.692649  2126.000000
weighted avg   0.709345  0.694262  0.691602  2126.000000
```

Experiment 1f / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4349
Accuracy: 0.7995
F1-score: 0.8078
Confusion matrix:
   0    1
0[5437 2107]
1[ 776 6058]

Training classification report:
              precision    recall  f1-score       support
class_0        0.875101  0.720705  0.790434   7544.000000
class_1        0.741947  0.886450  0.807787   6834.000000
accuracy       0.799485  0.799485  0.799485      0.799485
macro avg      0.808524  0.803578  0.799111  14378.000000
weighted avg   0.811812  0.799485  0.798682  14378.000000


Test metrics:
Loss:     0.5724
Accuracy: 0.7131
F1-score: 0.7291
Confusion matrix:
   0   1
0[695 418]
1[192 821]

Test classification report:
              precision    recall  f1-score      support
class_0        0.783540  0.624438  0.695000  1113.000000
class_1        0.662631  0.810464  0.729130  1013.000000
accuracy       0.713076  0.713076  0.713076     0.713076
macro avg      0.723086  0.717451  0.712065  2126.000000
weighted avg   0.725929  0.713076  0.711262  2126.000000
```

Experiment 1f / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5575
Accuracy: 0.7147
F1-score: 0.7493
Confusion matrix:
   0    1
0[4147 3397]
0[ 705 6129]

Training classification report:
              precision    recall  f1-score       support
class_0        0.854699  0.549708  0.669087   7544.000000
class_1        0.643397  0.896839  0.749267   6834.000000
accuracy       0.714703  0.714703  0.714703      0.714703
macro avg      0.749048  0.723274  0.709177  14378.000000
weighted avg   0.754265  0.714703  0.707197  14378.000000


Test metrics:
Loss:     0.6055
Accuracy: 0.6853
F1-score: 0.7275
Confusion matrix:
   0   1
0[564 549]
1[120 893]

Test classification report:
              precision    recall  f1-score      support
class_0        0.824561  0.506739  0.627713  1113.000000
class_1        0.619279  0.881540  0.727495  1013.000000
accuracy       0.685325  0.685325  0.685325     0.685325
macro avg      0.721920  0.694139  0.677604  2126.000000
weighted avg   0.726748  0.685325  0.675257  2126.000000
```

Experiment 2a / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5491
Accuracy: 0.7265
F1-score: 0.7413
Confusion matrix:
    0      1
0[14439  8194]
1[ 3602 16900]

Training classification report:
              precision    recall  f1-score       support
class_0        0.800344  0.637962  0.709987  22633.000000
class_1        0.673468  0.824310  0.741293  20502.000000
accuracy       0.726533  0.726533  0.726533      0.726533
macro avg      0.736906  0.731136  0.725640  43135.000000
weighted avg   0.740040  0.726533  0.724867  43135.000000


Test metrics:
Loss:     0.5963
Accuracy: 0.6971
F1-score: 0.7229
Confusion matrix:
   0     1
0[1926 1413]
1[ 519 2520]

Test classification report:
              precision    recall  f1-score      support
class_0        0.787730  0.576819  0.665975  3339.000000
class_1        0.640732  0.829220  0.722892  3039.000000
accuracy       0.697084  0.697084  0.697084     0.697084
macro avg      0.714231  0.703020  0.694433  6378.000000
weighted avg   0.717688  0.697084  0.693095  6378.000000
```

Experiment 2a / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4545
Accuracy: 0.7865
F1-score: 0.7874
Confusion matrix:
   0      1
0[16871  5762]
1[ 3446 17056]

Training classification report:
              precision    recall  f1-score       support
class_0        0.830388  0.745416  0.785611  22633.000000
class_1        0.747480  0.831919  0.787442  20502.000000
accuracy       0.786531  0.786531  0.786531      0.786531
macro avg      0.788934  0.788667  0.786527  43135.000000
weighted avg   0.790982  0.786531  0.786482  43135.000000


Test metrics:
Loss:     0.5485
Accuracy: 0.7154
F1-score: 0.7245
Confusion matrix:
   0     1
0[2176 1163]
1[ 652 2387]

Test classification report:
              precision    recall  f1-score      support
class_0        0.769448  0.651692  0.705692  3339.000000
class_1        0.672394  0.785456  0.724541  3039.000000
accuracy       0.715428  0.715428  0.715428     0.715428
macro avg      0.720921  0.718574  0.715116  6378.000000
weighted avg   0.723204  0.715428  0.714673  6378.000000
```

Experiment 2a / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5501
Accuracy: 0.7218
F1-score: 0.7594
Confusion matrix:
   0      1
0[12195 10438]
1[ 1562 18940]

Training classification report:
              precision    recall  f1-score       support
class_0        0.886458  0.538815  0.670239  22633.000000
class_1        0.644700  0.923812  0.759423  20502.000000
accuracy       0.721804  0.721804  0.721804      0.721804
macro avg      0.765579  0.731314  0.714831  43135.000000
weighted avg   0.771551  0.721804  0.712628  43135.000000



Test metrics:
Loss:     0.6162
Accuracy: 0.6860
F1-score: 0.7331
Confusion matrix:
   0    1
0[1624 1715]
1[ 288 2751]

Test classification report:
              precision    recall  f1-score      support
class_0        0.849372  0.486373  0.618549  3339.000000
class_1        0.615987  0.905232  0.733111  3039.000000
accuracy       0.685952  0.685952  0.685952     0.685952
macro avg      0.732680  0.695803  0.675830  6378.000000
weighted avg   0.738169  0.685952  0.673136  6378.000000
```

Experiment 2b / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5186
Accuracy: 0.7531
F1-score: 0.7783
Confusion matrix:
   0      1
0[13796  8837]
1[ 1812 18690]

Training classification report:
              precision    recall  f1-score       support
class_0        0.883906  0.609552  0.721529  22633.000000
class_1        0.678970  0.911618  0.778280  20502.000000
accuracy       0.753124  0.753124  0.753124      0.753124
macro avg      0.781438  0.760585  0.749905  43135.000000
weighted avg   0.786500  0.753124  0.748503  43135.000000


Test metrics:
Loss:     0.6568
Accuracy: 0.6918
F1-score: 0.7305
Confusion matrix:
   0    1
0[1748 1591]
1[ 375 2664]

Test classification report:
              precision    recall  f1-score      support
class_0        0.823363  0.523510  0.640059  3339.000000
class_1        0.626087  0.876604  0.730463  3039.000000
accuracy       0.691753  0.691753  0.691753     0.691753
macro avg      0.724725  0.700057  0.685261  6378.000000
weighted avg   0.729365  0.691753  0.683135  6378.000000
```

Experiment 2b / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4201
Accuracy: 0.8126
F1-score: 0.8028
Confusion matrix:
   0      1
0[18603  4030]
1[ 4052 16450]

Training classification report:
              precision    recall  f1-score       support
class_0        0.821143  0.821941  0.821542  22633.000000
class_1        0.803223  0.802361  0.802791  20502.000000
accuracy       0.812635  0.812635  0.812635      0.812635
macro avg      0.812183  0.812151  0.812167  43135.000000
weighted avg   0.812626  0.812635  0.812630  43135.000000


Test metrics:
Loss:     0.5586
Accuracy: 0.7167
F1-score: 0.7106
Confusion matrix:
   0     1
0[2352  987]
1[ 820 2219]

Test classification report:
              precision    recall  f1-score      support
class_0        0.741488  0.704403  0.722470  3339.000000
class_1        0.692140  0.730174  0.710649  3039.000000
accuracy       0.716682  0.716682  0.716682     0.716682
macro avg      0.716814  0.717288  0.716559  6378.000000
weighted avg   0.717974  0.716682  0.716837  6378.000000
```

Experiment 2b / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5165
Accuracy: 0.7453
F1-score: 0.7704
Confusion matrix:
   0      1
0[13715  8918]
1[ 2070 18432]

Training classification report:
              precision    recall  f1-score       support
class_0        0.868863  0.605974  0.713988  22633.000000
class_1        0.673931  0.899034  0.770375  20502.000000
accuracy       0.745265  0.745265  0.745265      0.745265
macro avg      0.771397  0.752504  0.742182  43135.000000
weighted avg   0.776212  0.745265  0.740789  43135.000000


Test metrics:
Loss:     0.5856
Accuracy: 0.7032
F1-score: 0.7384
Confusion matrix:
  0     1
0[1814 1525]
1[ 368 2671]

Test classification report:
              precision    recall  f1-score      support
class_0        0.831347  0.543276  0.657127  3339.000000
class_1        0.636559  0.878908  0.738355  3039.000000
accuracy       0.703198  0.703198  0.703198     0.703198
macro avg      0.733953  0.711092  0.697741  6378.000000
weighted avg   0.738534  0.703198  0.695831  6378.000000
```

Experiment 2c / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4865
Accuracy: 0.7665
F1-score: 0.7843
Confusion matrix:
   0      1
0[14741  7892]
1[ 2182 18320]

Training classification report:
              precision    recall  f1-score       support
class_0        0.871063  0.651306  0.745323  22633.000000
class_1        0.698917  0.893571  0.784347  20502.000000
accuracy       0.766454  0.766454  0.766454      0.766454
macro avg      0.784990  0.772438  0.764835  43135.000000
weighted avg   0.789242  0.766454  0.763871  43135.000000


Test metrics:
Loss:     0.6561
Accuracy: 0.6916
F1-score: 0.7245
Confusion matrix:
   0    1
0[1824 1515]
1[ 452 2587]

Test classification report:
              precision    recall  f1-score      support
class_0        0.801406  0.546271  0.649688  3339.000000
class_1        0.630668  0.851267  0.724548  3039.000000
accuracy       0.691596  0.691596  0.691596     0.691596
macro avg      0.716037  0.698769  0.687118  6378.000000
weighted avg   0.720052  0.691596  0.685358  6378.000000
```

Experiment 2c / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4267
Accuracy: 0.8078
F1-score: 0.7997
Confusion matrix:
   0      1
0[18299  4334]
1[ 3955 16547]

Training classification report:
              precision    recall  f1-score       support
class_0        0.822279  0.808510  0.815336  22633.000000
class_1        0.792443  0.807092  0.799700  20502.000000
accuracy       0.807836  0.807836  0.807836      0.807836
macro avg      0.807361  0.807801  0.807518  43135.000000
weighted avg   0.808098  0.807836  0.807905  43135.000000


Test metrics:
Loss:     0.5788
Accuracy: 0.7107
F1-score: 0.7045
Confusion matrix:
   0    1
0[2334 1005]
1[ 840 2199]

Test classification report:
              precision    recall  f1-score      support
class_0        0.735350  0.699012  0.716720  3339.000000
class_1        0.686330  0.723593  0.704469  3039.000000
accuracy       0.710724  0.710724  0.710724     0.710724
macro avg      0.710840  0.711302  0.710595  6378.000000
weighted avg   0.711993  0.710724  0.710883  6378.000000
```

Experiment 2c / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5341
Accuracy: 0.7322
F1-score: 0.7625
Confusion matrix:
   0      1
0[13035  9598]
1[ 1954 18548]

Training classification report:
              precision    recall  f1-score      support
class_0        0.869638  0.575929  0.692946  22633.00000
class_1        0.658992  0.904692  0.762539  20502.00000
accuracy       0.732190  0.732190  0.732190      0.73219
macro avg      0.764315  0.740311  0.727742  43135.00000
weighted avg   0.769518  0.732190  0.726023  43135.00000



Test metrics:
Loss:     0.6016
Accuracy: 0.6936
F1-score: 0.7318
Confusion matrix:
   0    1
0[1758 1581]
1[ 373 2666]

Test classification report:
              precision    recall  f1-score      support
class_0        0.824965  0.526505  0.642779  3339.000000
class_1        0.627737  0.877262  0.731814  3039.000000
accuracy       0.693634  0.693634  0.693634     0.693634
macro avg      0.726351  0.701884  0.687297  6378.000000
weighted avg   0.730989  0.693634  0.685203  6378.000000
```

Experiment 2d / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4689
Accuracy: 0.7772
F1-score: 0.7968
Confusion matrix:
   0      1
0[14682  7951]
1[ 1660 18842]

Training classification report:
              precision    recall  f1-score       support
class_0        0.898421  0.648699  0.753406  22633.000000
class_1        0.703243  0.919032  0.796786  20502.000000
accuracy       0.777188  0.777188  0.777188      0.777188
macro avg      0.800832  0.783866  0.775096  43135.000000
weighted avg   0.805654  0.777188  0.774025  43135.000000



Test metrics:
Loss:     0.6673
Accuracy: 0.6866
F1-score: 0.7213
Confusion matrix:
   0    1
0[1792 1547]
1[ 452 2587]

Test classification report:
              precision    recall  f1-score      support
class_0        0.798574  0.536688  0.641949  3339.000000
class_1        0.625786  0.851267  0.721316  3039.000000
accuracy       0.686579  0.686579  0.686579     0.686579
macro avg      0.712180  0.693977  0.681632  6378.000000
weighted avg   0.716244  0.686579  0.679766  6378.000000
```

Experiment 2d / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4454
Accuracy: 0.7952
F1-score: 0.7791
Confusion matrix:
   0     1
0[18726  3907]
1[ 4925 15577]

Training classification report:
              precision    recall  f1-score       support
class_0        0.791764  0.827376  0.809178  22633.000000
class_1        0.799476  0.759780  0.779123  20502.000000
accuracy       0.795247  0.795247  0.795247      0.795247
macro avg      0.795620  0.793578  0.794150  43135.000000
weighted avg   0.795430  0.795247  0.794893  43135.000000


Test metrics:
Loss:     0.5976
Accuracy: 0.7093
F1-score: 0.6936
Confusion matrix:
   0     1
0[2426  913]
1[ 941 2098]

Test classification report:
              precision    recall  f1-score      support
class_0        0.720523  0.726565  0.723531  3339.000000
class_1        0.696778  0.690359  0.693554  3039.000000
accuracy       0.709313  0.709313  0.709313     0.709313
macro avg      0.708651  0.708462  0.708542  6378.000000
weighted avg   0.709209  0.709313  0.709247  6378.000000
```

Experiment 2d / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4989
Accuracy: 0.7586
F1-score: 0.7723
Confusion matrix:
   0      1
0[15065  7568]
1[ 2844 17658]

Training classification report:
              precision    recall  f1-score       support
class_0        0.841197  0.665621  0.743180  22633.000000
class_1        0.699992  0.861282  0.772306  20502.000000
accuracy       0.758618  0.758618  0.758618      0.758618
macro avg      0.770595  0.763451  0.757743  43135.000000
weighted avg   0.774083  0.758618  0.757023  43135.000000


Test metrics:
Loss:     0.5798
Accuracy: 0.7040
F1-score: 0.7253
Confusion matrix:
  0     1
0[1997 1342]
1[ 546 2493]

Test classification report:
              precision    recall  f1-score      support
class_0        0.785293  0.598083  0.679021  3339.000000
class_1        0.650065  0.820336  0.725342  3039.000000
accuracy       0.703982  0.703982  0.703982     0.703982
macro avg      0.717679  0.709209  0.702181  6378.000000
weighted avg   0.720859  0.703982  0.701092  6378.000000
```

Experiment 2e / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.6279
Accuracy: 0.6744
F1-score: 0.7290
Confusion matrix:
   0     1
0[10195 12438]
1[ 1608 18894]

Training classification report:
              precision    recall  f1-score       support
class_0        0.863763  0.450448  0.592113  22633.000000
class_1        0.603026  0.921569  0.729020  20502.000000
accuracy       0.674371  0.674371  0.674371      0.674371
macro avg      0.733395  0.686009  0.660566  43135.000000
weighted avg   0.739835  0.674371  0.657184  43135.000000


Test metrics:
Loss:     0.6997
Accuracy: 0.6510
F1-score: 0.7134
Confusion matrix:
  0     1
0[1381 1958]
1[ 268 2771]

Test classification report:
              precision    recall  f1-score      support
class_0        0.837477  0.413597  0.553729  3339.000000
class_1        0.585959  0.911813  0.713440  3039.000000
accuracy       0.650988  0.650988  0.650988     0.650988
macro avg      0.711718  0.662705  0.633584  6378.000000
weighted avg   0.717633  0.650988  0.629828  6378.000000
```

Experiment 2e / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5102
Accuracy: 0.7515
F1-score: 0.7318
Confusion matrix:
   0      1
0[17797  4836]
1[ 5882 14620]

Training classification report:
              precision    recall  f1-score       support
class_0        0.751594  0.786330  0.768570  22633.000000
class_1        0.751439  0.713101  0.731768  20502.000000
accuracy       0.751524  0.751524  0.751524      0.751524
macro avg      0.751517  0.749715  0.750169  43135.000000
weighted avg   0.751521  0.751524  0.751078  43135.000000


Test metrics:
Loss:     0.5877
Accuracy: 0.6965
F1-score: 0.6772
Confusion matrix:
  0     1
0[2411  928]
1[1008 2031]

Test classification report:
              precision    recall  f1-score      support
class_0        0.705177  0.722072  0.713525  3339.000000
class_1        0.686381  0.668312  0.677226  3039.000000
accuracy       0.696457  0.696457  0.696457     0.696457
macro avg      0.695779  0.695192  0.695375  6378.000000
weighted avg   0.696221  0.696457  0.696229  6378.000000
```

Experiment 2e / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.5108
Accuracy: 0.7507
F1-score: 0.7487
Confusion matrix:
   0      1
0[16362  6271]
1[ 4483 16019]

Training classification report:
              precision    recall  f1-score      support
class_0        0.784936  0.722927  0.752657  22633.00000
class_1        0.718663  0.781338  0.748691  20502.00000
accuracy       0.750690  0.750690  0.750690      0.75069
macro avg      0.751800  0.752133  0.750674  43135.00000
weighted avg   0.753437  0.750690  0.750772  43135.00000


Test metrics:
Loss:     0.5923
Accuracy: 0.6875
F1-score: 0.6916
Confusion matrix:
   0    1
0[2150 1189]
1[ 804 2235]

Test classification report:
              precision    recall  f1-score     support
class_0        0.727827  0.643905  0.683299  3339.00000
class_1        0.652745  0.735439  0.691629  3039.00000
accuracy       0.687520  0.687520  0.687520     0.68752
macro avg      0.690286  0.689672  0.687464  6378.00000
weighted avg   0.692052  0.687520  0.687268  6378.00000
```

Experiment 2f / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4772
Accuracy: 0.7665
F1-score: 0.7873
Confusion matrix:
   0      1
0[14427  8206]
1[ 1864 18638]

Training classification report:
              precision    recall  f1-score       support
class_0        0.885581  0.637432  0.741291  22633.000000
class_1        0.694308  0.909082  0.787310  20502.000000
accuracy       0.766547  0.766547  0.766547      0.766547
macro avg      0.789944  0.773257  0.764301  43135.000000
weighted avg   0.794669  0.766547  0.763164  43135.000000


Test metrics:
Loss:     0.6017
Accuracy: 0.6872
F1-score: 0.7224
Confusion matrix:
  0     1
0[1787 1552]
1[ 443 2596]

Test classification report:
              precision    recall  f1-score      support
class_0        0.801345  0.535190  0.641767  3339.000000
class_1        0.625844  0.854228  0.722415  3039.000000
accuracy       0.687206  0.687206  0.687206     0.687206
macro avg      0.713595  0.694709  0.682091  6378.000000
weighted avg   0.717722  0.687206  0.680194  6378.000000
```

Experiment 2f / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4265
Accuracy: 0.8054
F1-score: 0.8051
Confusion matrix:
   0      1
0[17408  5225]
1[ 3167 17335]

Training classification report:
              precision    recall  f1-score       support
class_0        0.846075  0.769142  0.805777  22633.000000
class_1        0.768395  0.845527  0.805118  20502.000000
accuracy       0.805448  0.805448  0.805448      0.805448
macro avg      0.807235  0.807335  0.805447  43135.000000
weighted avg   0.809154  0.805448  0.805464  43135.000000


Test metrics:
Loss:     0.5875
Accuracy: 0.7035
F1-score: 0.7092
Confusion matrix:
   0    1
0[2181 1158]
1[ 733 2306]

Test classification report:
              precision    recall  f1-score      support
class_0        0.748456  0.653190  0.697585  3339.000000
class_1        0.665704  0.758802  0.709211  3039.000000
accuracy       0.703512  0.703512  0.703512     0.703512
macro avg      0.707080  0.705996  0.703398  6378.000000
weighted avg   0.709026  0.703512  0.703125  6378.000000
```

Experiment 2f / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.3933
Accuracy: 0.8228
F1-score: 0.8178
Confusion matrix:
   0      1
0[18342  4291]
1[ 3351 17151]

Training classification report:
              precision    recall  f1-score       support
class_0        0.845526  0.810410  0.827596  22633.000000
class_1        0.799879  0.836553  0.817805  20502.000000
accuracy       0.822835  0.822835  0.822835      0.822835
macro avg      0.822702  0.823481  0.822700  43135.000000
weighted avg   0.823830  0.822835  0.822942  43135.000000


Test metrics:
Loss:     0.6062
Accuracy: 0.7059
F1-score: 0.7074
Confusion matrix:
   0     1
0[2234 1105]
1[ 771 2268]

Test classification report:
              precision    recall  f1-score      support
class_0        0.743428  0.669063  0.704288  3339.000000
class_1        0.672398  0.746298  0.707424  3039.000000
accuracy       0.705864  0.705864  0.705864     0.705864
macro avg      0.707913  0.707680  0.705856  6378.000000
weighted avg   0.709584  0.705864  0.705782  6378.000000
```

Experiment 3a / `Random_state: 42`

```
0='Bagdad unrequited', 1='SR', 2='The quick brown fox', 3='all_other', 4='childish
piecemeal', 5='fulfilling Dylks', 6='gunpowder exclaim', 7='husbandman Jacques', 8=
'jumps over the lazy', 9='microscope dusky', 10='overpower lightsome', 11='recogniz
e Boxer', 12= 'smart squid who asks'


Training metrics:
Loss:     0.6361
Accuracy: 0.6096
F1-score: 0.9258
Confusion matrix:
      0    1    2    3    4    5    6    7    8    9   10   11   12
0 [ 114    0    1    0    0    0    1    0    0    0    0    1    0]
1 [   0 4337    0 2496    0    1    0    0    0    0    0    0    0]
2 [   0    0  124    0    0    0    0    0    0    0    0    0    0]
3 [   0 3617    0 3927    0    0    0    0    0    0    0    0    0]
4 [   2    0    0    0  114    0    0    0    1    0    0    0    0]
5 [   0    0    0    0    0  116    0    0    0    0    0    0    0]
6 [   0    0    0    0    0    0  120    0    0    0    0    0    0]
7 [   0    0    2    0    0    0    0  115    0    0    0    0    0]
8 [   0    0    0    0    0    0    0    0  122    0    0    0    0]
9 [   0    0    0    0    0    0    0    0    0  123    0    1    0]
10[   0    0    0    0    0    0    0    0    0    0  120    2    0]
11[   0    0    0    0    0    0    0    0    0    0    0  118    0]
12[   0    0    0    0    0    0    0    0    0    1    0    1  119]

Training classification report:
              precision    recall  f1-score       support
class_0        0.982759  0.974359  0.978541    117.000000
class_1        0.545260  0.634621  0.586557   6834.000000
class_2        0.976378  1.000000  0.988048    124.000000
class_3        0.611397  0.520546  0.562325   7544.000000
class_4        1.000000  0.974359  0.987013    117.000000
class_5        0.991453  1.000000  0.995708    116.000000
class_6        0.991736  1.000000  0.995851    120.000000
class_7        1.000000  0.982906  0.991379    117.000000
class_8        0.991870  1.000000  0.995918    122.000000
class_9        0.991935  0.991935  0.991935    124.000000
class_10       1.000000  0.983607  0.991736    122.000000
class_11       0.959350  1.000000  0.979253    118.000000
class_12       1.000000  0.983471  0.991667    121.000000
accuracy       0.609646  0.609646  0.609646      0.609646
macro avg      0.926318  0.926600  0.925841  15696.000000
weighted avg   0.614358  0.609646  0.608769  15696.000000
```

```
Test metrics:
Loss:     0.7126
Accuracy: 0.5715
F1-score: 0.7529
Confusion matrix:
      0   1   2   3   4   5   6   7   8   9  10  11  12
0 [  9   0   2   0   0   0   1   0   1   0   0   2   1]
1 [  0 619   0 394   0   0   0   0   0   0   0   0   0]
2 [  0   0  12   0   0   0   0   0   2   1   0   0   1]
3 [  0 555   0 558   0   0   0   0   0   0   0   0   0]
4 [  1   0   1   0  11   0   0   0   2   0   1   0   0]
5 [  0   0   0   0   0  12   0   0   0   1   0   1   1]
6 [  0   0   0   0   0   0  15   0   0   0   0   1   0]
7 [  0   0   1   0   1   0   0  13   0   0   0   1   0]
8 [  0   0   0   0   0   0   0   1  14   1   0   0   0]
9 [  0   0   0   0   0   0   0   0   1  15   0   0   0]
10[  0   0   0   0   0   0   0   0   2   1  11   1   1]
11[  0   0   0   0   0   0   0   0   0   0   0  16   0]
12[  0   0   0   0   0   0   0   0   0   5   0   1  10]

Test classification report:
              precision    recall  f1-score      support
class_0        0.900000  0.562500  0.692308    16.000000
class_1        0.527257  0.611056  0.566072  1013.000000
class_2        0.750000  0.750000  0.750000    16.000000
class_3        0.586134  0.501348  0.540436  1113.000000
class_4        0.916667  0.687500  0.785714    16.000000
class_5        1.000000  0.800000  0.888889    15.000000
class_6        0.937500  0.937500  0.937500    16.000000
class_7        0.928571  0.812500  0.866667    16.000000
class_8        0.636364  0.875000  0.736842    16.000000
class_9        0.625000  0.937500  0.750000    16.000000
class_10       0.916667  0.687500  0.785714    16.000000
class_11       0.695652  1.000000  0.820513    16.000000
class_12       0.714286  0.625000  0.666667    16.000000
accuracy       0.571491  0.571491  0.571491     0.571491
macro avg      0.779546  0.752877  0.752871  2301.000000
weighted avg   0.577927  0.571491  0.570596  2301.000000
```

Experiment 3a / `Random_state: 31`

```
0='Bagdad unrequited', 1='SR', 2='The quick brown fox', 3='all_other', 4='childish
piecemeal', 5='fulfilling Dylks', 6='gunpowder exclaim', 7='husbandman Jacques', 8=
'jumps over the lazy', 9='microscope dusky', 10='overpower lightsome', 11='recogniz
e Boxer', 12= 'smart squid who asks'

Training metrics:
Loss:     0.4562
Accuracy: 0.7824
F1-score: 0.9544
Confusion matrix:
      0    1    2    3    4    5    6    7    8    9   10   11   12
0 [ 117    0    0    0    0    0    0    0    0    0    0    0    0]
1 [   1 5261    0 1572    0    0    0    0    0    0    0    0    0]
2 [   0    0  124    0    0    0    0    0    0    0    0    0    0]
3 [   2 1825    2 5711    0    2    0    0    1    0    1    0    0]
4 [   0    0    0    0  116    1    0    0    0    0    0    0    0]
5 [   0    0    0    0    0  116    0    0    0    0    0    0    0]
6 [   0    0    1    0    0    0  119    0    0    0    0    0    0]
7 [   0    0    1    0    0    0    0  116    0    0    0    0    0]
8 [   0    0    1    0    0    0    0    0  121    0    0    0    0]
9 [   0    0    3    0    0    0    0    0    0  121    0    0    0]
10[   1    0    0    0    0    0    0    0    0    0  121    0    0]
11[   0    0    0    0    0    0    0    0    0    0    0  118    0]
12[   0    0    1    0    0    0    0    1    0    0    0    0  119]

Training classification report:
              precision    recall  f1-score       support
class_0        0.966942  1.000000  0.983193    117.000000
class_1        0.742450  0.769827  0.755891   6834.000000
class_2        0.932331  1.000000  0.964981    124.000000
class_3        0.784155  0.757025  0.770351   7544.000000
class_4        1.000000  0.991453  0.995708    117.000000
class_5        0.974790  1.000000  0.987234    116.000000
class_6        1.000000  0.991667  0.995816    120.000000
class_7        0.991453  0.991453  0.991453    117.000000
class_8        0.991803  0.991803  0.991803    122.000000
class_9        1.000000  0.975806  0.987755    124.000000
class_10       0.991803  0.991803  0.991803    122.000000
class_11       1.000000  1.000000  1.000000    118.000000
class_12       1.000000  0.983471  0.991667    121.000000
accuracy       0.782365  0.782365  0.782365      0.782365
macro avg      0.951979  0.957255  0.954435  15696.000000
weighted avg   0.782963  0.782365  0.782427  15696.000000
```

```
Test metrics:
Loss:     0.5997
Accuracy: 0.7071
F1-score: 0.7616
Confusion matrix:
     0   1   2   3   4   5   6   7   8   9  10  11  12
0 [ 14   0   1   0   0   0   0   0   0   1   0   0   0]
1 [  0 731   1 281   0   0   0   0   0   0   0   0   0]
2 [  0   0  16   0   0   0   0   0   0   0   0   0   0]
3 [  0 351   0 760   0   0   0   0   0   0   2   0   0]
4 [  0   0   1   0  13   1   0   0   1   0   0   0   0]
5 [  0   0   1   1   0  13   0   0   0   0   0   0   0]
6 [  0   0   1   0   0   1  10   2   1   1   0   0   0]
7 [  0   0   4   0   0   0   0  11   0   0   0   0   1]
8 [  1   0   0   0   0   0   0   0  14   0   0   0   1]
9 [  3   0   1   0   2   1   0   1   0   7   1   0   0]
10[  0   0   0   1   0   0   1   0   0   0  14   0   0]
11[  0   0   1   0   0   2   0   0   0   1   0  12   0]
12[  0   0   0   0   0   3   0   1   0   0   0   0  12]

Test classification report:
              precision    recall  f1-score      support
class_0        0.777778  0.875000  0.823529    16.000000
class_1        0.675601  0.721619  0.697852  1013.000000
class_2        0.592593  1.000000  0.744186    16.000000
class_3        0.728667  0.682839  0.705009  1113.000000
class_4        0.866667  0.812500  0.838710    16.000000
class_5        0.619048  0.866667  0.722222    15.000000
class_6        0.909091  0.625000  0.740741    16.000000
class_7        0.733333  0.687500  0.709677    16.000000
class_8        0.875000  0.875000  0.875000    16.000000
class_9        0.700000  0.437500  0.538462    16.000000
class_10       0.823529  0.875000  0.848485    16.000000
class_11       1.000000  0.750000  0.857143    16.000000
class_12       0.857143  0.750000  0.800000    16.000000
accuracy       0.707084  0.707084  0.707084     0.707084
macro avg      0.781419  0.766048  0.761617  2301.000000
weighted avg   0.710490  0.707084  0.707018  2301.000000
```

Experiment 3a / `Random_state: 7`

```
0='Bagdad unrequited', 1='SR', 2='The quick brown fox', 3='all_other', 4='childish
piecemeal', 5='fulfilling Dylks', 6='gunpowder exclaim', 7='husbandman Jacques', 8=
'jumps over the lazy', 9='microscope dusky', 10='overpower lightsome', 11='recogniz
e Boxer', 12= 'smart squid who asks'

Training metrics:
Loss:     0.4487
Accuracy: 0.7901
F1-score: 0.9517
Confusion matrix:
      0    1    2    3    4    5    6    7    8    9   10   11   12
0 [ 117    0    0    0    0    0    0    0    0    0    0    0    0]
1 [   1 5657    0 1171    0    1    1    0    2    0    0    1    0]
2 [   0    0  124    0    0    0    0    0    0    0    0    0    0]
3 [   0 2082    0 5426    1    8    4   11    2    5    1    2    2]
4 [   0    0    0    0  117    0    0    0    0    0    0    0    0]
5 [   0    0    0    0    0  116    0    0    0    0    0    0    0]
6 [   0    0    0    0    0    0  120    0    0    0    0    0    0]
7 [   0    0    0    0    0    0    0  117    0    0    0    0    0]
8 [   0    0    0    0    0    0    0    0  122    0    0    0    0]
9 [   0    0    0    0    0    0    0    0    0  124    0    0    0]
10[   0    0    0    0    0    0    0    0    0    0  122    0    0]
11[   0    0    0    0    0    0    0    0    0    0    0  118    0]
12[   0    0    0    0    0    0    0    0    0    0    0    0  121]

Training classification report:
              precision    recall  f1-score       support
class_0        0.991525  1.000000  0.995745    117.000000
class_1        0.730973  0.827773  0.776367   6834.000000
class_2        1.000000  1.000000  1.000000    124.000000
class_3        0.822495  0.719247  0.767414   7544.000000
class_4        0.991525  1.000000  0.995745    117.000000
class_5        0.928000  1.000000  0.962656    116.000000
class_6        0.960000  1.000000  0.979592    120.000000
class_7        0.914062  1.000000  0.955102    117.000000
class_8        0.968254  1.000000  0.983871    122.000000
class_9        0.961240  1.000000  0.980237    124.000000
class_10       0.991870  1.000000  0.995918    122.000000
class_11       0.975207  1.000000  0.987448    118.000000
class_12       0.983740  1.000000  0.991803    121.000000
accuracy       0.790074  0.790074  0.790074      0.790074
macro avg      0.939915  0.965155  0.951684  15696.000000
weighted avg   0.795019  0.790074  0.789542  15696.000000
```

```
Test metrics:
Loss:     0.6118
Accuracy: 0.7101
F1-score: 0.7491
Confusion matrix:
     0   1   2   3   4   5   6   7   8   9  10  11  12
0 [ 13   0   0   0   1   0   0   0   0   1   0   0   1]
1 [  0 792   0 216   1   1   0   1   0   1   0   1   0]
2 [  3   0   9   0   1   0   0   1   1   0   0   0   1]
3 [  0 400   1 705   0   4   0   2   0   0   0   1   0]
4 [  0   0   0   0  13   0   0   0   1   1   0   0   1]
5 [  0   0   0   0   0  12   1   0   0   0   1   0   1]
6 [  0   0   0   0   0   0   9   3   3   0   0   0   1]
7 [  0   0   1   0   0   0   0  15   0   0   0   0   0]
8 [  0   0   0   0   0   0   0   1  14   0   0   0   1]
9 [  0   0   0   0   0   0   0   1   0  14   0   0   1]
10[  0   0   1   0   0   1   0   1   0   0  13   0   0]
11[  0   0   0   0   0   0   0   1   1   2   0  12   0]
12[  1   1   0   0   0   0   0   1   0   0   0   0  13]

Test classification report:
              precision    recall  f1-score      support
class_0        0.764706  0.812500  0.787879    16.000000
class_1        0.663873  0.781836  0.718042  1013.000000
class_2        0.750000  0.562500  0.642857    16.000000
class_3        0.765472  0.633423  0.693215  1113.000000
class_4        0.812500  0.812500  0.812500    16.000000
class_5        0.666667  0.800000  0.727273    15.000000
class_6        0.900000  0.562500  0.692308    16.000000
class_7        0.555556  0.937500  0.697674    16.000000
class_8        0.700000  0.875000  0.777778    16.000000
class_9        0.736842  0.875000  0.800000    16.000000
class_10       0.928571  0.812500  0.866667    16.000000
class_11       0.857143  0.750000  0.800000    16.000000
class_12       0.650000  0.812500  0.722222    16.000000
accuracy       0.710126  0.710126  0.710126     0.710126
macro avg      0.750102  0.771366  0.749109  2301.000000
weighted avg   0.720104  0.710126  0.709010  2301.000000
```

Experiment 3b / `Random_state: 42`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4724
Accuracy: 0.7748
F1-score: 0.7847
Confusion matrix:
   0    1
0[5239 2305]
1[ 933 5901]

Training classification report:
              precision    recall  f1-score       support
class_0        0.848833  0.694459  0.763925   7544.000000
class_1        0.719108  0.863477  0.784707   6834.000000
accuracy       0.774795  0.774795  0.774795      0.774795
macro avg      0.783971  0.778968  0.774316  14378.000000
weighted avg   0.787174  0.774795  0.773803  14378.000000

Validation metrics:
Loss:     0.5303
Accuracy: 0.7292
F1-score: 0.7429
Confusion matrix:
   0     1
0[1620  895]
1[ 403 1875]


Test classification report:
              precision    recall  f1-score      support
class_0        0.792627  0.618149  0.694599  1113.000000
class_1        0.662162  0.822310  0.733598  1013.000000
accuracy       0.715428  0.715428  0.715428     0.715428
macro avg      0.727394  0.720230  0.714098  2126.000000
weighted avg   0.730463  0.715428  0.713181  2126.000000
```

Experiment 3b / `Random_state: 31`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4354
Accuracy: 0.7949
F1-score: 0.7743
Confusion matrix:
   0    1
0[6370 1174]
1[1775 5059]

Training classification report:
              precision    recall  f1-score       support
class_0        0.782075  0.844380  0.812034   7544.000000
class_1        0.811648  0.740269  0.774317   6834.000000
accuracy       0.794895  0.794895  0.794895      0.794895
macro avg      0.796861  0.792324  0.793175  14378.000000
weighted avg   0.796131  0.794895  0.794107  14378.000000

Validation metrics:
Loss:     0.5243
Accuracy: 0.7421
F1-score: 0.7130
Confusion matrix:
   0     1
0[2022  493]
1[ 743 1535]


Test classification report:
              precision    recall  f1-score      support
class_0        0.720459  0.734052  0.727192  1113.000000
class_1        0.701613  0.687068  0.694264  1013.000000
accuracy       0.711665  0.711665  0.711665     0.711665
macro avg      0.711036  0.710560  0.710728  2126.000000
weighted avg   0.711479  0.711665  0.711502  2126.000000
```

Experiment 3b / `Random_state: 7`

```
0='all_other', 1='SR'

Training metrics:
Loss:     0.4929
Accuracy: 0.7623
F1-score: 0.7686
Confusion matrix:
   0    1
0[5285 2259]
1[1158 5676]

Training classification report:
              precision    recall  f1-score       support
class_0        0.820270  0.700557  0.755702   7544.000000
class_1        0.715312  0.830553  0.768637   6834.000000
accuracy       0.762345  0.762345  0.762345      0.762345
macro avg      0.767791  0.765555  0.762169  14378.000000
weighted avg   0.770382  0.762345  0.761850  14378.000000

Test metrics:
Loss:     0.5642
Accuracy: 0.7126
F1-score: 0.7261
Confusion matrix:
   0   1
0[705 408]
1[203 810]

Test classification report:
              precision    recall  f1-score      support
class_0        0.776432  0.633423  0.697674  1113.000000
class_1        0.665025  0.799605  0.726132  1013.000000
accuracy       0.712606  0.712606  0.712606     0.712606
macro avg      0.720728  0.716514  0.711903  2126.000000
weighted avg   0.723348  0.712606  0.711234  2126.000000
```

Experiment 4a / `Random_state: 42`

```
0='SR', 1='all_other', 2='binary_label' 3='single_label'


Training metrics:
Loss:     0.7006
Accuracy: 0.6966
F1-score: 0.6452
Confusion matrix:
     0    1    2    3
0[5529   92   28 1185]
1[  39  568   10  159]
2[ 105   60  470  341]
3[1581  330  153 2807]

Training classification report:
              precision    recall  f1-score       support
class_0        0.762200  0.809043  0.784923   6834.000000
class_1        0.540952  0.731959  0.622125    776.000000
class_2        0.711044  0.481557  0.574221    976.000000
class_3        0.624889  0.576268  0.599594   4871.000000
accuracy       0.696589  0.696589  0.696589      0.696589
macro avg      0.659771  0.649707  0.645216  13457.000000
weighted avg   0.696029  0.696589  0.693171  13457.000000


Test metrics:
Loss:     0.9323
Accuracy: 0.5903
F1-score: 0.4300
Confusion matrix:
    0   1   2   3
0[794  22   9 188]
1[ 11  48  14  40]
2[ 31  35  19  60]
3[304  70  33 316]

Test classification report:
              precision    recall  f1-score      support
class_0        0.696491  0.783810  0.737575  1013.000000
class_1        0.274286  0.424779  0.333333   113.000000
class_2        0.253333  0.131034  0.172727   145.000000
class_3        0.523179  0.437068  0.476262   723.000000
accuracy       0.590271  0.590271  0.590271     0.590271
macro avg      0.436822  0.444173  0.429975  1994.000000
weighted avg   0.577498  0.590271  0.578843  1994.000000
```

Experiment 4a / `Random_state: 31`

```
0='SR', 1='all_other', 2='binary_label' 3='single_label'

Training metrics:
Loss:     0.8823
Accuracy: 0.6200
F1-score: 0.5626
Confusion matrix:
     0    1    2    3
0[5074  212  588  960]
1[   8  612  139   17]
2[  31   65  842   38]
3[1310  474 1271 1816]

Training classification report:
              precision    recall  f1-score       support
class_0        0.789974  0.742464  0.765482   6834.000000
class_1        0.449010  0.788660  0.572230    776.000000
class_2        0.296479  0.862705  0.441300    976.000000
class_3        0.641469  0.372819  0.471566   4871.000000
accuracy       0.620049  0.620049  0.620049      0.620049
macro avg      0.544233  0.691662  0.562645  13457.000000
weighted avg   0.680766  0.620049  0.624438  13457.000000


Test metrics:
Loss:     1.0630
Accuracy: 0.5281
F1-score: 0.4062
Confusion matrix:
    0   1   2   3
0[743  59  92 119]
1[  7  50  44  12]
2[ 17  46  68  14]
3[259  96 176 192]

Test classification report:
              precision    recall  f1-score      support
class_0        0.724172  0.733465  0.728789  1013.000000
class_1        0.199203  0.442478  0.274725   113.000000
class_2        0.178947  0.468966  0.259048   145.000000
class_3        0.569733  0.265560  0.362264   723.000000
accuracy       0.528084  0.528084  0.528084     0.528084
macro avg      0.418014  0.477617  0.406206  1994.000000
weighted avg   0.598776  0.528084  0.536001  1994.000000
```

Experiment 4a / `Random_state: 7`

```
0='SR', 1='all_other', 2='binary_label' 3='single_label'

Training metrics:
Loss:     1.1634
Accuracy: 0.4841
F1-score: 0.4075
Confusion matrix:
     0    1    2    3
0[4383  454 1273  724]
1[  11  499  258    8]
2[  40  201  695   40]
3[1157  739 2037  938]

Training classification report:
              precision    recall  f1-score       support
class_0        0.783938  0.641352  0.705513   6834.000000
class_1        0.263603  0.643041  0.373923    776.000000
class_2        0.163031  0.712090  0.265318    976.000000
class_3        0.548538  0.192568  0.285063   4871.000000
accuracy       0.484135  0.484135  0.484135      0.484135
macro avg      0.439777  0.547263  0.407454  13457.000000
weighted avg   0.623693  0.484135  0.502276  13457.000000


Test metrics:
Loss:     1.2484
Accuracy: 0.4418
F1-score: 0.3396
Confusion matrix:
    0   1   2   3
0[652 112 155  94]
1[  3  61  43   6]
2[ 14  47  75   9]
3[208 163 259  93]

Test classification report:
              precision    recall  f1-score      support
class_0        0.743444  0.643633  0.689947  1013.000000
class_1        0.159269  0.539823  0.245968   113.000000
class_2        0.140977  0.517241  0.221566   145.000000
class_3        0.460396  0.128631  0.201081   723.000000
accuracy       0.441825  0.441825  0.441825     0.441825
macro avg      0.376021  0.457332  0.339640  1994.000000
weighted avg   0.563899  0.441825  0.453470  1994.000000
```

Experiment 4b / `Random_state: 42`

```
0='SR', 1='all_other', 2='form', 3='rhythm'



Training metrics:
Loss:     1.3328
Accuracy: 0.3417
F1-score: 0.3044
Confusion matrix:
     0    1    2    3
0[1280  237 4520  797]
1[  77  225 1214 1053]
2[ 240  185 2559  544]
3[  87   43  468  850]

Training classification report:
              precision    recall  f1-score       support
class_0        0.760095  0.187299  0.300540   6834.000000
class_1        0.326087  0.087583  0.138079   2569.000000
class_2        0.292090  0.725340  0.416470   3528.000000
class_3        0.262022  0.587017  0.362319   1448.000000
accuracy       0.341748  0.341748  0.341748      0.341748
macro avg      0.410074  0.396810  0.304352  14379.000000
weighted avg   0.517568  0.341748  0.306180  14379.000000


Test metrics:
Loss:     1.3527
Accuracy: 0.3222
F1-score: 0.2918
Confusion matrix:
    0   1   2   3
0[177  51 659 126]
1[ 14  34 179 141]
2[ 52  36 346  83]
3[ 19   8  73 128]

Test classification report:
              precision    recall  f1-score      support
class_0        0.675573  0.174729  0.277647  1013.000000
class_1        0.263566  0.092391  0.136821   368.000000
class_2        0.275259  0.669246  0.390079   517.000000
class_3        0.267782  0.561404  0.362606   228.000000
accuracy       0.322201  0.322201  0.322201     0.322201
macro avg      0.370545  0.374442  0.291788  2126.000000
weighted avg   0.463175  0.322201  0.289723  2126.000000
```

Experiment 4b / `Random_state: 31`

```
0='SR', 1='all_other', 2='form', 3='rhythm'

Training metrics:
Loss:     1.0647
Accuracy: 0.4973
F1-score: 0.4557
Confusion matrix:
     0    1    2    3
0[2543  545 3723   23]
1[  34 1687  799   49]
2[ 320  512 2689    7]
3[ 162  724  330  232]

Training classification report:
              precision    recall  f1-score       support
class_0        0.831317  0.372110  0.514101   6834.000000
class_1        0.486448  0.656676  0.558887   2569.000000
class_2        0.356584  0.762188  0.485861   3528.000000
class_3        0.745981  0.160221  0.263786   1448.000000
accuracy       0.497322  0.497322  0.497322      0.497322
macro avg      0.605082  0.487799  0.455659  14379.000000
weighted avg   0.644629  0.497322  0.489966  14379.000000


Test metrics:
Loss:     1.1644
Accuracy: 0.4520
F1-score: 0.3985
Confusion matrix:
    0   1   2   3
0[387  97 525   4]
1[  9 211 145   3]
2[ 66 112 339   0]
3[ 25 120  59  24]

Test classification report:
              precision    recall  f1-score      support
class_0        0.794661  0.382034  0.516000  1013.000000
class_1        0.390741  0.573370  0.464758   368.000000
class_2        0.317416  0.655706  0.427760   517.000000
class_3        0.774194  0.105263  0.185328   228.000000
accuracy       0.452023  0.452023  0.452023     0.452023
macro avg      0.569253  0.429093  0.398462  2126.000000
weighted avg   0.606493  0.452023  0.450210  2126.000000
```

Experiment 4b / `Random_state: 7`

```
0='SR', 1='all_other', 2='form', 3='rhythm'

Training metrics:
Loss:     0.8373
Accuracy: 0.6452
F1-score: 0.6123
Confusion matrix:
     0    1    2    3
0[4887   41 1802  104]
1[ 190  966  971  442]
2[ 867  122 2464   75]
3[ 270   67  150  961]

Training classification report:
              precision    recall  f1-score       support
class_0        0.786450  0.715101  0.749080   6834.000000
class_1        0.807692  0.376022  0.513147   2569.000000
class_2        0.457397  0.698413  0.552776   3528.000000
class_3        0.607459  0.663674  0.634323   1448.000000
accuracy       0.645247  0.645247  0.645247      0.645247
macro avg      0.664750  0.613302  0.612332  14379.000000
weighted avg   0.691485  0.645247  0.647207  14379.000000


Test metrics:
Loss:     1.0274
Accuracy: 0.5546
F1-score: 0.4927
Confusion matrix:
    0   1   2   3
0[706  17 268  22]
1[ 54  96 149  69]
2[199  32 269  17]
3[ 57  31  32 108]

Test classification report:
              precision    recall  f1-score      support
class_0        0.694882  0.696940  0.695909  1013.000000
class_1        0.545455  0.260870  0.352941   368.000000
class_2        0.374652  0.520309  0.435628   517.000000
class_3        0.500000  0.473684  0.486486   228.000000
accuracy       0.554563  0.554563  0.554563     0.554563
macro avg      0.528747  0.487951  0.492741  2126.000000
weighted avg   0.570243  0.554563  0.550789  2126.000000
```

Experiment 4c / `Random_state: 42`

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', 15='SR', 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', 20='all_other'

Training metrics:
Loss: 1.5605
Accuracy: 0.4322
F1-score: 0.5271
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	750	18	1	1	33	24	247	52	21	2	8	4	83	17	11	120	8	27	12	81	65
1	1	301	0	0	0	2	0	2	0	0	0	0	0	2	1	0	0	1	0	0	12
2	0	0	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	1	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	58	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
6	39	2	0	0	2	2	410	26	24	0	0	0	10	0	1	11	0	9	1	15	19
7	1	0	0	0	0	0	2	179	0	0	0	0	1	0	0	0	0	0	0	0	0
8	0	1	0	0	0	1	0	0	68	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	85	0	0	0	0	0	0	0	0	0	0	0
10	0	1	0	0	0	0	0	0	0	0	190	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	1	0	0	0	0	0	0	208	0	0	0	0	0	0	0	0	2
12	1	0	0	1	0	0	0	0	0	0	0	0	149	0	0	0	0	0	0	0	1
13	13	3	0	0	1	1	13	7	1	0	0	2	6	221	5	20	1	4	1	1	17
14	6	1	0	0	1	0	8	1	1	1	0	1	1	7	248	1	0	1	0	2	8
15	654	80	6	5	168	149	1086	296	220	35	18	23	490	420	115	1807	21	271	36	681	253
16	0	0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	239	1	0	0	3
18	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	95	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	50	0	0
20	6	0	0	0	3	2	16	1	0	0	0	0	1	0	0	10	0	2	0	229	0
21	149	452	2	3	44	15	335	129	40	13	21	183	54	28	84	58	163	94	17	37	752

Training classification report:

	precision	recall	f1-score	support
class_0	0.462677	0.473186	0.467873	1585.000000
class_1	0.350407	0.934783	0.509738	322.000000
class_2	0.800000	1.000000	0.888889	36.000000
class_3	0.807692	0.976744	0.884211	43.000000
class_4	0.277143	1.000000	0.434004	97.000000
class_5	0.228346	0.983051	0.370607	59.000000
class_6	0.193396	0.718039	0.304719	571.000000
class_7	0.257925	0.978142	0.408210	183.000000
class_8	0.181333	0.971429	0.305618	70.000000
class_9	0.625000	1.000000	0.769231	85.000000
class_10	0.801688	0.994764	0.887850	191.000000
class_11	0.494062	0.985782	0.658228	211.000000
class_12	0.187186	0.980263	0.314346	152.000000
class_13	0.317986	0.697161	0.436759	317.000000
class_14	0.533333	0.861111	0.658699	288.000000
class_15	0.891465	0.264413	0.407855	6834.000000
class_16	0.553241	0.971545	0.705015	246.000000
class_17	0.188119	0.989583	0.316140	96.000000
class_18	0.427350	1.000000	0.598802	50.000000
class_19	0.218929	0.848148	0.348024	270.000000
class_20	0.664311	0.281332	0.395269	2673.000000
accuracy	0.432158	0.432158	0.432158	0.432158
macro avg	0.450552	0.852832	0.527147	14379.000000
weighted avg	0.682686	0.432158	0.433347	14379.000000

Test metrics:
Loss: 2.2086
Accuracy: 0.2681
F1-score: 0.2008
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	46	4	3	0	10	8	34	11	8	0	1	1	24	4	1	29	2	10	3	7	13
1	1	30	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	0	0	0	15
2	0	0	0	0	0	0	0	1	0	0	0	0	0	0	2	0	0	0	0	0	1
3	0	0	0	0	1	0	0	1	0	1	0	0	0	0	0	0	0	1	0	0	1
4	0	0	0	0	1	0	1	1	0	0	0	0	2	0	0	0	0	1	0	0	1
5	0	0	0	0	0	3	1	0	0	0	0	0	2	1	0	0	1	0	0	0	1
6	7	0	0	0	1	1	39	13	7	0	0	1	5	1	0	4	0	6	1	4	11
7	3	0	0	0	1	0	7	3	0	0	0	0	3	0	0	0	0	4	0	1	5
8	0	0	0	0	0	0	2	2	1	0	0	0	1	0	0	1	0	1	0	0	0
9	0	0	0	0	0	0	1	1	0	3	0	0	0	3	0	0	0	0	0	1	5
10	2	3	0	0	0	0	0	1	0	0	12	1	0	0	0	0	0	0	0	0	7
11	1	2	0	0	0	0	0	1	0	2	2	15	0	3	0	0	1	1	0	0	7
12	2	0	0	0	2	2	3	0	1	0	0	0	7	0	0	5	0	0	0	0	0
13	2	0	0	1	0	0	0	0	2	2	0	0	3	17	1	12	0	1	0	3	5
14	1	1	0	0	1	0	4	1	2	0	0	0	0	2	23	0	0	1	0	5	3
15	70	14	2	0	28	21	133	37	56	8	1	6	96	68	19	244	0	55	4	108	43
16	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	31	0	0	0	8
17	1	0	0	0	0	0	3	2	0	0	0	0	1	0	1	1	0	3	0	2	0
18	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	1	0	2
19	3	0	0	0	0	2	7	1	4	1	0	0	2	3	2	6	1	0	0	7	0
20	17	64	1	0	6	4	42	24	11	8	4	32	12	10	12	9	25	17	3	5	84

Test classification report:

	precision	recall	f1-score	support
class_0	0.292994	0.210046	0.244681	219.000000
class_1	0.250000	0.588235	0.350877	51.000000
class_2	0.000000	0.000000	0.000000	4.000000
class_3	0.000000	0.000000	0.000000	5.000000
class_4	0.019608	0.142857	0.034483	7.000000
class_5	0.073171	0.333333	0.120000	9.000000
class_6	0.139785	0.386139	0.205263	101.000000
class_7	0.030000	0.111111	0.047244	27.000000
class_8	0.010753	0.125000	0.019802	8.000000
class_9	0.120000	0.214286	0.153846	14.000000
class_10	0.571429	0.461538	0.510638	26.000000
class_11	0.267857	0.428571	0.329670	35.000000
class_12	0.044304	0.318182	0.077778	22.000000
class_13	0.151786	0.346939	0.211180	49.000000
class_14	0.370968	0.522727	0.433962	44.000000
class_15	0.777070	0.240869	0.367747	1013.000000
class_16	0.500000	0.738095	0.596154	42.000000
class_17	0.029703	0.214286	0.052174	14.000000
class_18	0.083333	0.142857	0.105263	7.000000
class_19	0.048951	0.179487	0.076923	39.000000
class_20	0.396226	0.215385	0.279070	390.000000
accuracy	0.268109	0.268109	0.268109	0.268109
macro avg	0.198949	0.281902	0.200798	2126.000000
weighted avg	0.521628	0.268109	0.312305	2126.000000

Experiment 4c / `Random_state: 31`

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', **15='SR'**, 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', **20='all_other'**

Training metrics:
Loss: 2.4383
Accuracy: 0.1862
F1-score: 0.2194
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	354	34	42	474	28	103	38	122	15	10	51	18	16	3	46	55	4	22	57	77	16]
1 [	0	251	0	14	2	8	0	7	2	0	4	10	1	0	3	1	1	0	9	3	6]
2 [	2	0	20	7	0	3	0	1	0	0	0	0	0	0	1	1	0	0	0	1	0]
3 [	0	0	0	43	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
4 [	3	5	4	5	59	11	0	0	0	0	1	2	0	0	0	0	0	0	4	0	3]
5 [	2	0	0	1	1	45	0	5	4	0	0	0	0	0	0	1	0	0	0	0	0]
6 [	47	13	9	211	8	27	60	123	21	0	4	0	4	0	6	5	3	12	7	8	3]
7 [	1	10	3	51	1	4	6	82	2	1	0	3	1	0	2	0	2	5	1	5	3]
8 [	1	0	2	13	0	10	6	7	21	0	0	0	0	0	2	4	0	3	0	1	0]
9 [	1	3	1	5	1	11	1	21	3	22	0	4	1	0	1	0	0	4	0	2	4]
10[	2	3	0	5	0	0	0	0	0	0	178	1	1	0	1	0	0	0	0	0	0]
11[	0	9	1	6	1	3	0	14	0	4	8	143	0	0	2	0	2	3	2	1	12]
12[	11	0	8	23	9	32	2	10	4	3	1	1	23	0	4	7	0	5	3	4	2]
13[	7	8	16	34	8	21	4	39	13	10	1	5	2	27	18	19	2	6	3	69	5]
14[	4	2	12	9	1	6	2	8	5	0	4	4	0	1	206	5	0	1	7	9	2]
15[	331	152	326	1100	152	610	191	955	248	136	78	60	79	58	266	659	23	122	168	1080	40]
16[	0	6	0	0	0	3	0	5	0	0	0	5	1	0	0	0	222	1	0	0	3]
17[	0	0	2	17	0	4	5	25	0	3	1	0	0	0	1	1	0	34	0	1	2]
18[	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	48	0	0]
19[	9	3	6	36	3	16	4	44	3	1	3	0	1	3	13	15	0	2	6	101	1]
20[	103	591	20	381	37	101	63	293	51	17	73	284	17	3	103	26	229	68	98	36	79]

Training classification report:

	precision	recall	f1-score	support
class_0	0.403189	0.223344	0.287454	1585.000000
class_1	0.230275	0.779503	0.355524	322.000000
class_2	0.042373	0.555556	0.078740	36.000000
class_3	0.017652	1.000000	0.034691	43.000000
class_4	0.189711	0.608247	0.289216	97.000000
class_5	0.044204	0.762712	0.083565	59.000000
class_6	0.157068	0.105079	0.125918	571.000000
class_7	0.046564	0.448087	0.084362	183.000000
class_8	0.053571	0.300000	0.090909	70.000000
class_9	0.106280	0.258824	0.150685	85.000000
class_10	0.436275	0.931937	0.594324	191.000000
class_11	0.264815	0.677725	0.380826	211.000000
class_12	0.156463	0.151316	0.153846	152.000000
class_13	0.284211	0.085174	0.131068	317.000000
class_14	0.305185	0.715278	0.427830	288.000000
class_15	0.824781	0.096430	0.172671	6834.000000
class_16	0.454918	0.902439	0.604905	246.000000
class_17	0.118056	0.354167	0.177083	96.000000
class_18	0.116223	0.960000	0.207343	50.000000
class_19	0.072246	0.374074	0.121103	270.000000
class_20	0.436464	0.029555	0.055361	2673.000000
accuracy	0.186174	0.186174	0.186174	0.186174
macro avg	0.226692	0.491402	0.219401	14379.000000
weighted avg	0.566120	0.186174	0.183102	14379.000000

Test metrics:
Loss: 2.6367
Accuracy: 0.1341
F1-score: 0.1439
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	20	6	9	37	6	37	6	27	10	1	5	2	9	1	4	10	1	6	3	17	2
1	2	34	0	1	0	3	0	1	0	0	3	2	0	0	0	0	1	0	3	0	1
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	0	0	1
3	0	1	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
4	0	1	0	1	1	1	0	2	0	0	0	0	0	0	1	0	0	0	0	0	0
5	0	1	0	1	0	4	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0
6	4	3	5	26	3	8	11	21	4	0	0	1	0	0	1	2	1	6	2	2	1
7	0	0	0	10	1	1	2	10	1	0	0	1	0	0	0	0	0	0	0	0	1
8	0	0	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	3	0	0	0
9	0	0	0	2	1	2	0	2	0	1	1	1	0	0	0	1	1	0	0	1	1
10	0	3	0	1	0	0	0	0	0	0	18	3	0	0	0	0	0	1	0	0	0
11	0	0	1	5	0	2	0	1	0	2	3	15	0	0	0	0	1	2	0	1	2
12	0	0	2	5	0	4	1	1	3	0	0	0	1	0	0	2	0	0	1	2	0
13	2	0	3	4	0	5	0	5	1	3	1	0	0	3	4	1	1	1	0	15	0
14	1	0	3	4	1	0	0	4	2	0	0	1	0	0	24	0	0	1	0	3	0
15	34	33	75	141	30	116	28	131	49	19	9	10	11	8	45	86	2	22	13	146	5
16	0	2	0	0	0	0	0	2	0	0	0	2	0	0	0	0	33	1	0	0	2
17	0	0	1	5	0	1	1	6	0	0	0	0	0	0	0	0	0	0	0	0	0
18	1	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	3	0	0
19	0	1	3	6	0	5	0	8	2	1	0	0	0	0	1	1	1	0	0	10	0
20	13	77	4	53	2	29	5	63	8	1	6	47	3	0	20	1	29	7	8	6	8

Test classification report:

	precision	recall	f1-score	support
class_0	0.259740	0.091324	0.135135	219.000000
class_1	0.208589	0.666667	0.317757	51.000000
class_2	0.000000	0.000000	0.000000	4.000000
class_3	0.009772	0.600000	0.019231	5.000000
class_4	0.022222	0.142857	0.038462	7.000000
class_5	0.018349	0.444444	0.035242	9.000000
class_6	0.200000	0.108911	0.141026	101.000000
class_7	0.034965	0.370370	0.063898	27.000000
class_8	0.000000	0.000000	0.000000	8.000000
class_9	0.035714	0.071429	0.047619	14.000000
class_10	0.391304	0.692308	0.500000	26.000000
class_11	0.176471	0.428571	0.250000	35.000000
class_12	0.041667	0.045455	0.043478	22.000000
class_13	0.250000	0.061224	0.098361	49.000000
class_14	0.235294	0.545455	0.328767	44.000000
class_15	0.811321	0.084896	0.153709	1013.000000
class_16	0.458333	0.785714	0.578947	42.000000
class_17	0.000000	0.000000	0.000000	14.000000
class_18	0.090909	0.428571	0.150000	7.000000
class_19	0.049261	0.256410	0.082645	39.000000
class_20	0.333333	0.020513	0.038647	390.000000
accuracy	0.134055	0.134055	0.134055	0.134055
macro avg	0.172726	0.278339	0.143949	2126.000000
weighted avg	0.518852	0.134055	0.143217	2126.000000

Experiment 4c / `Random_state: 7`

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', **15='SR'**, 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', **20='all_other'**

Training metrics:
Loss: 1.9267
Accuracy: 0.2798
F1-score: 0.3747
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	787	51	17	5	54	48	77	8	1	28	3	17	75	93	81	41	8	3	54	113	21]
1 [	0	279	1	0	7	0	0	0	0	3	1	4	0	2	7	2	3	0	9	1	3]
2 [	0	0	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
3 [	0	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0]
4 [	0	0	0	0	97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
5 [	0	0	0	0	0	59	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
6 [	131	19	18	1	9	20	164	15	0	20	0	2	19	13	36	22	3	4	20	48	7]
7 [	18	16	1	0	2	3	21	61	0	5	0	2	1	7	10	10	0	0	7	11	8]
8 [	2	0	0	0	0	2	1	2	41	0	0	0	13	1	5	1	0	0	1	1	0]
9 [	0	0	0	0	0	1	0	0	0	84	0	0	0	0	0	0	0	0	0	0	0]
10[	1	1	0	0	2	1	0	0	0	0	179	1	0	0	3	0	0	0	3	0	0]
11[	3	12	0	0	2	0	0	0	0	2	0	180	1	1	4	1	2	0	1	0	2]
12[	9	0	0	0	5	7	1	0	0	1	0	1	104	13	6	3	0	0	0	2	0]
13[	27	5	1	1	3	8	4	2	0	8	1	2	4	207	14	14	3	0	5	5	3]
14[	1	3	0	0	0	0	0	0	0	0	0	3	0	2	277	0	0	0	1	1	0]
15[	978	165	108	21	260	250	265	72	10	231	10	60	319	1608	563	753	23	25	183	877	53]
16[	0	2	0	0	0	1	0	0	0	0	0	2	0	1	0	0	237	0	0	1	2]
17[	6	0	13	0	2	2	2	1	0	0	0	0	1	0	0	4	0	56	0	8	1]
18[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	50	0	0]
19[	20	0	1	0	9	5	6	2	0	2	0	0	4	16	22	8	1	0	2	171	1]
20[	268	716	30	15	90	44	133	63	3	82	13	284	46	65	202	42	230	12	80	96	159]

Training classification report:

	precision	recall	f1-score	support
class_0	0.349622	0.496530	0.410323	1585.000000
class_1	0.219858	0.866460	0.350723	322.000000
class_2	0.159292	1.000000	0.274809	36.000000
class_3	0.494118	0.976744	0.656250	43.000000
class_4	0.178967	1.000000	0.303599	97.000000
class_5	0.130820	1.000000	0.231373	59.000000
class_6	0.243323	0.287215	0.263454	571.000000
class_7	0.269912	0.333333	0.298289	183.000000
class_8	0.745455	0.585714	0.656000	70.000000
class_9	0.180258	0.988235	0.304900	85.000000
class_10	0.864734	0.937173	0.899497	191.000000
class_11	0.322581	0.853081	0.468140	211.000000
class_12	0.177172	0.684211	0.281461	152.000000
class_13	0.102021	0.652997	0.176471	317.000000
class_14	0.225203	0.961806	0.364954	288.000000
class_15	0.835738	0.110184	0.194699	6834.000000
class_16	0.464706	0.963415	0.626984	246.000000
class_17	0.560000	0.583333	0.571429	96.000000
class_18	0.119904	1.000000	0.214133	50.000000
class_19	0.128090	0.633333	0.213084	270.000000
class_20	0.611538	0.059484	0.108421	2673.000000
accuracy	0.279783	0.279783	0.279783	0.279783
macro avg	0.351586	0.713012	0.374714	14379.000000
weighted avg	0.615128	0.279783	0.242958	14379.000000

Test metrics:
Loss: 2.3356
Accuracy: 0.1877
F1-score: 0.1659
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	64	8	6	0	9	16	12	2	0	8	1	1	18	28	13	7	2	1	3	15	5
1	1	33	0	0	1	1	0	0	0	1	1	2	0	0	0	1	1	0	2	2	5
2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0
3	0	0	0	0	1	0	0	1	1	0	0	0	0	2	0	0	0	0	0	0	0
4	2	1	0	0	2	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
5	0	1	0	0	1	4	0	0	0	0	0	0	0	2	0	0	1	0	0	0	0
6	17	4	5	0	4	3	19	9	1	8	1	1	5	0	10	5	1	0	1	4	3
7	4	3	1	0	1	1	6	3	0	1	0	1	1	0	1	0	0	0	0	3	1
8	0	0	2	0	0	0	1	0	0	3	0	0	0	0	0	1	0	0	0	1	0
9	2	1	0	1	0	1	0	0	0	3	0	0	1	2	0	0	0	0	0	1	2
10	0	5	0	1	0	0	0	0	0	0	11	4	0	0	1	0	0	0	2	0	2
11	1	2	0	0	0	1	0	0	0	3	1	21	0	1	1	0	2	0	0	1	1
12	5	0	0	0	1	1	2	0	1	1	0	0	6	1	1	2	0	0	1	0	0
13	4	1	0	0	0	2	2	0	0	0	0	1	3	24	4	2	1	0	0	5	0
14	3	0	0	0	1	0	1	0	0	0	1	0	0	3	31	0	0	0	0	2	2
15	105	27	23	7	43	46	27	11	5	37	1	7	66	259	80	109	1	2	22	121	14
16	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	36	0	0	1	2
17	2	0	2	0	0	0	2	0	0	1	0	0	1	1	2	0	0	0	0	3	0
18	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	1
19	5	0	2	0	0	2	0	1	0	1	0	0	3	5	6	2	1	0	0	11	0
20	28	101	4	1	18	5	25	13	0	16	1	39	10	12	33	9	30	0	8	18	19

Test classification report:

	precision	recall	f1-score	support
class_0	0.260163	0.292237	0.275269	219.000000
class_1	0.173684	0.647059	0.273859	51.000000
class_2	0.000000	0.000000	0.000000	4.000000
class_3	0.000000	0.000000	0.000000	5.000000
class_4	0.024390	0.285714	0.044944	7.000000
class_5	0.048193	0.444444	0.086957	9.000000
class_6	0.195876	0.188119	0.191919	101.000000
class_7	0.075000	0.111111	0.089552	27.000000
class_8	0.000000	0.000000	0.000000	8.000000
class_9	0.035714	0.214286	0.061224	14.000000
class_10	0.611111	0.423077	0.500000	26.000000
class_11	0.272727	0.600000	0.375000	35.000000
class_12	0.052632	0.272727	0.088235	22.000000
class_13	0.070588	0.489796	0.123393	49.000000
class_14	0.166667	0.704545	0.269565	44.000000
class_15	0.789855	0.107601	0.189401	1013.000000
class_16	0.473684	0.857143	0.610169	42.000000
class_17	0.000000	0.000000	0.000000	14.000000
class_18	0.071429	0.428571	0.122449	7.000000
class_19	0.058511	0.282051	0.096916	39.000000
class_20	0.333333	0.048718	0.085011	390.000000
accuracy	0.187676	0.187676	0.187676	0.187676
macro avg	0.176836	0.304629	0.165898	2126.000000
weighted avg	0.507493	0.187676	0.187799	2126.000000

Experiment 5 / `Random_state: 42`

```
Training metrics:
Loss:                 0.5823
Binary accuracy:      0.8252
Exact-match accuracy: 0.5643
Macro F1-score:       0.6648

Training label-wise confusion matrices:

SR:
      TN   TP
TN [4726 2818]
TP [ 782 6052]

ABQRS:
      TN     TP
TN [11222   902]
TP [ 1176  1078]

AFIB:
       TN    TP
TN [12451   906]
TP [    9  1012]

all_other:
      TN   TP
TN [7054 1829]
TP [1632 3863]

0='SR', 1='ABQRS', 2='AFIB', 3='all_other_alone', 4='ABQRS + all_other', 5='AFIB +
all_other', 6='ABQRS + AFIB', 7='ABQRS + AFIB + all_other' 8='invalid_combination'

Training combination-class confusion matrix:
     0    1    2    3    4    5    6    7    8
0[5447  130   15  400   66   63   22   15  676]
1[ 591  323    5  150  192   23   33   24  244]
2[   4    1   14    2    0  248    6   38    9]
3[1310   69    7 1710  202  425   14   84  648]
4[  57   26    2  123  131   39   13   53   25]
5[   0    0    3    2    0  437    1   54    2]
6[   0    0    5    0    0   54   16   66    1]
7[   0    0    0    0    0   22    0   36    0]
8[   0    0    0    0    0    0    0    0    0]

Training multi-label classification report:
              precision    recall  f1-score  support
SR             0.682300  0.885572  0.770759   6834.0
ABQRS          0.544444  0.478261  0.509211   2254.0
AFIB           0.527633  0.991185  0.688670   1021.0
all_other      0.678672  0.703003  0.690623   5495.0
micro avg      0.650325  0.769354  0.704850  15604.0
macro avg      0.608262  0.764505  0.664816  15604.0
weighted avg   0.650989  0.769354  0.699387  15604.0
samples avg    0.676473  0.765174  0.703707  15604.0
```

```
Training combination-class classification report:
                           precision    recall  f1-score       support
SR                          0.735187  0.797044  0.764867   6834.000000
ABQRS                       0.588342  0.203785  0.302718   1585.000000
AFIB                        0.274510  0.043478  0.075067    322.000000
all_other_alone             0.716380  0.382636  0.498833   4469.000000
ABQRS + all_other           0.221658  0.279318  0.247170    469.000000
AFIB + all_other            0.333333  0.875752  0.482873    499.000000
ABQRS + AFIB                0.152381  0.112676  0.129555    142.000000
ABQRS + AFIB + all_other    0.097297  0.620690  0.168224     58.000000
invalid_combination         0.000000  0.000000  0.000000      0.000000
accuracy                    0.564334  0.564334  0.564334      0.564334
macro avg                   0.346565  0.368375  0.296590  14378.000000
weighted avg                0.663810  0.564334  0.580428  14378.000000


Test metrics:
Loss:                 0.7827
Binary accuracy:      0.7954
Exact-match accuracy: 0.5160
Macro F1-score:       0.5858

Test label-wise confusion matrices:

SR:
     TN  TP
TN [629 484]
TP [144 869]

ABQRS:
      TN   TP
TN [1696  108]
TP [ 233   89]

AFIB:
      TN   TP
TN [1822  152]
TP [  14  138]

all_other:
     TN  TP
TN [980 337]
TP [268 541]

Test combination-class confusion matrix:
    0   1   2   3   4   5   6   7   8
0[773  14   3  76  18  18   1   3 107]
1[118  17   0  30  13  10   1   4  26]
2[  2   0   0   2   1  35   0   6   5]
3[218   6   3 246  27  66   3   9 106]
4[ 13   5   0  12   4  10   1   8   5]
5[  0   0   2   0   1  50   0   2   1]
6[  1   2   0   0   1  10   4  16   0]
7[  0   0   0   2   0   5   1   3   0]
8[  0   0   0   0   0   0   0   0   0]
```

Test multi-label classification report:

	precision	recall	f1-score	support
SR	0.642276	0.857848	0.734573	1013.0
ABQRS	0.451777	0.276398	0.342967	322.0
AFIB	0.475862	0.907895	0.624434	152.0
all_other	0.616173	0.668727	0.641375	809.0
micro avg	0.602281	0.712979	0.652972	2296.0
macro avg	0.546522	0.677717	0.585837	2296.0
weighted avg	0.595345	0.712979	0.639523	2296.0
samples avg	0.627469	0.711508	0.652995	2296.0

Test combination-class classification report:

	precision	recall	f1-score	support
SR	0.687111	0.763080	0.723106	1013.000000
ABQRS	0.386364	0.077626	0.129278	219.000000
AFIB	0.000000	0.000000	0.000000	51.000000
all_other_alone	0.668478	0.359649	0.467681	684.000000
ABQRS + all_other	0.061538	0.068966	0.065041	58.000000
AFIB + all_other	0.245098	0.892857	0.384615	56.000000
ABQRS + AFIB	0.363636	0.117647	0.177778	34.000000
ABQRS + AFIB + all_other	0.058824	0.272727	0.096774	11.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.515992	0.515992	0.515992	0.515992
macro avg	0.274561	0.283617	0.227141	2126.000000
weighted avg	0.596520	0.515992	0.523580	2126.000000

Experiment 5 / `Random_state: 31`

```
Training metrics:
Loss:                 0.4992
Binary accuracy:      0.8135
Exact-match accuracy: 0.5118
Macro F1-score:       0.7072

Training label-wise confusion matrices:

SR:
      TN   TP
TN [6003 1541]
TP [1618 5216]

ABQRS:
      TN   TP
TN [8590 3534]
TP [ 275 1979]

AFIB:
       TN    TP
TN [12983   374]
TP [   19  1002]

all_other:
      TN   TP
TN [6778 2105]
TP [1262 4233]

0='SR', 1='ABQRS', 2='AFIB', 3='all_other_alone', 4='ABQRS + all_other', 5='AFIB +
all_other', 6='ABQRS + AFIB', 7='ABQRS + AFIB + all_other' 8='invalid_combination'

Training combination-class confusion matrix:
     0    1    2    3    4    5    6    7    8
0[3996  568    7  598  334   23   13   14 1281]
1[ 148  809    1   42  248    4   20   15  298]
2[   2    0    2    3    1  159   22  133    0]
3[ 636  310    1 1869  947  141   10   82  473]
4[   9   81    0   42  281    3    2   29   22]
5[   0    0    0    4    4  328    6  157    0]
6[   0    1    0    0    2   11   30   98    0]
7[   0    0    0    1    1    6    6   44    0]
8[   0    0    0    0    0    0    0    0    0]

Training multi-label classification report:
              precision    recall  f1-score  support
SR             0.771940  0.763243  0.767567   6834.0
ABQRS          0.358970  0.877995  0.509592   2254.0
AFIB           0.728198  0.981391  0.836045   1021.0
all_other      0.667876  0.770337  0.715457   5495.0
micro avg      0.621998  0.796591  0.698550  15604.0
macro avg      0.631746  0.848241  0.707165  15604.0
weighted avg   0.672778  0.796591  0.716432  15604.0
samples avg    0.658100  0.786758  0.698945  15604.0
```

```
Training combination-class classification report:
                           precision    recall  f1-score       support
SR                          0.834064  0.584723  0.687484   6834.000000
ABQRS                       0.457321  0.510410  0.482409   1585.000000
AFIB                        0.181818  0.006211  0.012012    322.000000
all_other_alone             0.730363  0.418214  0.531873   4469.000000
ABQRS + all_other           0.154565  0.599147  0.245737    469.000000
AFIB + all_other            0.485926  0.657315  0.558773    499.000000
ABQRS + AFIB                0.275229  0.211268  0.239044    142.000000
ABQRS + AFIB + all_other    0.076923  0.758621  0.139683     58.000000
invalid_combination         0.000000  0.000000  0.000000      0.000000
accuracy                    0.511824  0.511824  0.511824      0.511824
macro avg                   0.355134  0.416212  0.321890  14378.000000
weighted avg                0.702872  0.511824  0.575867  14378.000000


Test metrics:
Loss:                 0.7113
Binary accuracy:      0.7787
Exact-match accuracy: 0.4586
Macro F1-score:       0.6243

Test label-wise confusion matrices:

SR:
     TN  TP
TN [808 305]
TP [260 753]

ABQRS:
      TN   TP
TN [1355  449]
TP [ 147  175]

AFIB:
      TN   TP
TN [1897   77]
TP [  17  135]

all_other:
     TN  TP
TN [895 422]
TP[205 604]

Test combination-class confusion matrix:
    0   1   2   3   4   5   6   7   8
0[600  61   0 108  66   7   0   7 164]
1[ 67  47   0  29  31   4   3   1  37]
2[  1   0   0   2   4  23   2  19   0]
3[130  39   1 269 134  29   2  13  67]
4[  2   7   1  15  15   2   1   6   9]
5[  0   0   1   2   0  33   1  19   0]
6[  1   1   0   0   4   2   5  21   0]
7[  0   0   0   1   1   3   0   6   0]
8[  0   0   0   0   0   0   0   0   0]
```

Test multi-label classification report:

	precision	recall	f1-score	support
SR	0.711720	0.743337	0.727185	1013.0
ABQRS	0.280449	0.543478	0.369979	322.0
AFIB	0.636792	0.888158	0.741758	152.0
all_other	0.588694	0.746601	0.658311	809.0
micro avg	0.570890	0.726045	0.639187	2296.0
macro avg	0.554414	0.730393	0.624308	2296.0
weighted avg	0.602928	0.726045	0.653786	2296.0
samples avg	0.602540	0.718721	0.638523	2296.0

Test combination-class classification report:

	precision	recall	f1-score	support
SR	0.749064	0.592300	0.661521	1013.000000
ABQRS	0.303226	0.214612	0.251337	219.000000
AFIB	0.000000	0.000000	0.000000	51.000000
all_other_alone	0.631455	0.393275	0.484685	684.000000
ABQRS + all_other	0.058824	0.258621	0.095847	58.000000
AFIB + all_other	0.320388	0.589286	0.415094	56.000000
ABQRS + AFIB	0.357143	0.147059	0.208333	34.000000
ABQRS + AFIB + all_other	0.065217	0.545455	0.116505	11.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.458608	0.458608	0.458608	0.458608
macro avg	0.276146	0.304512	0.248147	2126.000000
weighted avg	0.607402	0.458608	0.514514	2126.000000

Experiment 5 / `Random_state: 7`

```
Training metrics:
Loss:                 0.4874
Binary accuracy:      0.8314
Exact-match accuracy: 0.5649
Macro F1-score:       0.6833

Training label-wise confusion matrices:

SR:
      TN   TP
TN [5599 1945]
TP [1112 5722]

ABQRS:
      TN   TP
TN [9970 2154]
TP [ 422 1832]

AFIB:
       TN    TP
TN [12117  1240]
TP [    0  1021]

all_other:
      TN   TP
TN [7651 1232]
TP [1591 3904]

0='SR', 1='ABQRS', 2='AFIB', 3='all_other_alone', 4='ABQRS + all_other', 5='AFIB +
all_other', 6='ABQRS + AFIB', 7='ABQRS + AFIB + all_other' 8='invalid_combination'

Training combination-class confusion matrix:
     0    1    2    3    4    5    6    7    8
0[4933  314   73  342  114   72   53   25  908]
1[ 250  729   12   33  132    7   83   26  313]
2[   0    0   39    0    0  121  103   58    1]
3[ 963  169   41 1758  410  461   46  153  468]
4[  22   64    3   45  208   14   27   70   16]
5[   0    0   13    0    0  334   41  111    0]
6[   0    0    0    0    0    8  100   34    0]
7[   0    0    0    0    0    8   29   21    0]
8[   0    0    0    0    0    0    0    0    0]

Training multi-label classification report:
              precision    recall  f1-score  support
SR             0.746315  0.837284  0.789187   6834.0
ABQRS          0.459609  0.812777  0.587179   2254.0
AFIB           0.451570  1.000000  0.622182   1021.0
all_other      0.760125  0.710464  0.734456   5495.0
micro avg      0.655066  0.799731  0.720205  15604.0
macro avg      0.604405  0.840131  0.683251  15604.0
weighted avg   0.690478  0.799731  0.729806  15604.0
samples avg    0.688981  0.794072  0.721735  15604.0
```

Training combination-class classification report:

	precision	recall	f1-score	support
SR	0.799773	0.721832	0.758806	6834.000000
ABQRS	0.571317	0.459937	0.509612	1585.000000
AFIB	0.215470	0.121118	0.155070	322.000000
all_other_alone	0.807163	0.393377	0.528960	4469.000000
ABQRS + all_other	0.240741	0.443497	0.312078	469.000000
AFIB + all_other	0.325854	0.669339	0.438320	499.000000
ABQRS + AFIB	0.207469	0.704225	0.320513	142.000000
ABQRS + AFIB + all_other	0.042169	0.362069	0.075540	58.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.564891	0.564891	0.564891	0.564891
macro avg	0.356662	0.430599	0.344322	14378.000000
weighted avg	0.720211	0.564891	0.613594	14378.000000

Test metrics:

Loss:	0.7440
Binary accuracy:	0.7793
Exact-match accuracy:	0.4854
Macro F1-score:	0.5755

Test label-wise confusion matrices:

SR:

	TN	TP
TN	730	383
TP	199	814

ABQRS:

	TN	TP
TN	1504	300
TP	179	143

AFIB:

	TN	TP
TN	1756	218
TP	9	143

all_other:

	TN	TP
TN	1027	290
TP	299	510

Test combination-class confusion matrix:

	0	1	2	3	4	5	6	7	8
0	717	41	12	80	23	10	10	11	109
1	90	34	2	25	21	5	9	3	30
2	2	0	4	1	2	22	11	9	0
3	187	30	8	216	45	75	14	30	79
4	6	5	1	12	7	8	6	7	6
5	0	1	3	1	0	35	3	13	0
6	1	0	0	0	0	8	14	11	0
7	0	0	0	1	0	4	1	5	0
8	0	0	0	0	0	0	0	0	0

Test multi-label classification report:

	precision	recall	f1-score	support
SR	0.680033	0.803554	0.736652	1013.0
ABQRS	0.322799	0.444099	0.373856	322.0
AFIB	0.396122	0.940789	0.557505	152.0
all_other	0.637500	0.630408	0.633934	809.0
micro avg	0.574795	0.701220	0.631744	2296.0
macro avg	0.509114	0.704713	0.575487	2296.0
weighted avg	0.596151	0.701220	0.637719	2296.0
samples avg	0.602854	0.696221	0.631217	2296.0

Test combination-class classification report:

	precision	recall	f1-score	support
SR	0.714855	0.707799	0.711310	1013.000000
ABQRS	0.306306	0.155251	0.206061	219.000000
AFIB	0.133333	0.078431	0.098765	51.000000
all_other_alone	0.642857	0.315789	0.423529	684.000000
ABQRS + all_other	0.071429	0.120690	0.089744	58.000000
AFIB + all_other	0.209581	0.625000	0.313901	56.000000
ABQRS + AFIB	0.205882	0.411765	0.274510	34.000000
ABQRS + AFIB + all_other	0.056180	0.454545	0.100000	11.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.485419	0.485419	0.485419	0.485419
macro avg	0.260047	0.318808	0.246424	2126.000000
weighted avg	0.593246	0.485419	0.514408	2126.000000